General Feedback Training: Learning by Matching Feedback Distributions

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Abstract

We formulate *General Feedback Training* (GFT) as matching the distribution of feedback elicited by an agent's executions to a desired distribution. The population formulation is agnostic to feedback modality and to whether observations are intended as descriptions, evaluations, or corrections: it does not need to interpret these communicative intentions to derive the update. GFT learns a feedback-conditioned execution posterior and distills its desired-feedback-conditioned mixture into a request-only policy. For finite feedback spaces and deterministic sequential environments, calibrated feedback, coverage, and exact population fitting yield an $O(1/T)$ last-iterate feedback-KL rate. A statistical separation condition implies $O(T^{-1/2})$ behavioral error even when individual updates reduce task success; a stronger common-optimal-feedback condition ensures monotonic success. We construct a separating language-feedback channel using truthful partial requirement audits and relate the exact population operator to FCP. For sampled batch GFT, an unrestricted finite-tabular empirical-MLE specialization has a high-probability behavioral guarantee, whereas a compact, posterior-realizable linear-softmax specialization approaches constrained stationary points under controlled optimization errors and increasing batches. These finite-space theorems do not establish convergence for arbitrary neural training or general continuous feedback. Distillation is motivated by rich, task-specific feedback available in training but not at deployment.

Keywords: general feedback training; feedback-distribution matching; language feedback; conditional imitation; policy distillation; population convergence.

1 Introduction and scope

Suppose a user asks an assistant to send a particular report to Alice and give Bob view-only access. The assistant may send the wrong report, or grant edit permission. Afterward, an observer may say "The recipient was right, but the file version and permissions were wrong." That feedback is informative, but it is not a new executable request and it does not provide the correct action sequence. Direct hindsight-request relabeling would therefore be inappropriate.

Our starting point is an explicit *feedback-distribution matching* objective: choose a policy whose executions elicit feedback distributed like feedback associated with desirable behavior. *General Feedback Training* (GFT) treats hindsight feedback as *evidence about which executions occurred*, rather than automatically as a reward, demonstration, or corrective instruction to follow. It learns the conditional distribution of past executions given the original request and resulting feedback, samples this conditional model using independently drawn desired feedback, and distills the new executions into a request-only policy. This is useful when ideal feedback is expensive, long, or task-specific and cannot readily be supplied for a new test request.

Modality and communicative intention. A feedback observation might be a verbal critique, a categorical assessment, a structured requirements report, or an annotated past trajectory. The

population mechanism does not need to know its modality or whether an individual message is intended descriptively, evaluatively, or correctively: it uses the statistical relationship between feedback and the execution that elicited it. For example, "you should have sent the newer proposal" can be used as an observation without being executed as an instruction. This does *not* mean that communicative intentions are irrelevant to constructing the actual feedback channel or specifying desired feedback, and it does not prevent neural models from exploiting linguistic meaning. Our proofs assume finite feedback alphabets; extending the same guarantees to general continuous images, audio, or other media requires additional assumptions.

The population theory has two logically distinct targets. The algorithm first converges in *feedback space*: would an external teacher describe its executions as it would describe an optimal execution? To conclude *behavioral optimality*, the feedback must make it impossible for an inappropriate mixture of executions to mimic the ideal feedback distribution. We develop a quantitative, constructive sufficient condition for this second step, rather than presupposing optimality identification.

Convergence is not a per-round improvement guarantee. GFT does not include a reward-based policy-improvement operator. Even under the optimality-separation condition used for its convergence theorem, a single ideal-feedback-conditioned distillation update can reduce task success. The proof instead follows a decreasing *feedback-distribution* KL potential and a comparator KL potential; Section 6.3 gives an explicit example satisfying the theorem's assumptions.

Relation to existing approaches. Hindsight experience replay [2] and hindsight-instruction relabeling [3, 4] use achieved outcomes or inferred instructions to relabel training examples. GFT instead keeps the original request fixed and conditions a separate execution model on *post-execution evaluative descriptions*; this distinction matters when a comment such as "permission was too broad" is not itself a valid replacement instruction. Distribution matching for categorical attributes of generated text is also studied in MaD-RL [7]; answer-level distributional alignment with marginalized latent reasoning appears in [8]. These are relevant distribution-level precedents, but neither directly formulates this fixed hindsight-feedback channel and its feedback-conditioned EM/distillation mechanism. The exact population operator below is also a latent-variable likelihood/EM update [1]; we do not claim that its multiplicative form or distribution matching in general is new. Feedback-Conditional Policy (FCP) [5] already learns a feedback posterior and repeats desired-feedback-conditioned online bootstrapping. Its idealized rollout marginal obeys the same exact population operator as GFT, though FCP does not separately distill a request-only student. Section 10 formalizes this equivalence, distinguishes feedback matching from task improvement, and explains when privileged training-time feedback makes request-only deployment valuable. Retrospective self-distillation (SDPO) [6] also uses rich execution feedback, but re-prompts the current model with *actual* feedback and distills its next-token probabilities, rather than sampling *ideal* feedback and cloning new conditional executions. Section 11 makes this distinction precise, including a deliberately qualified one-step population comparison. Our explicit *desired-versus-induced feedback KL* formulation supplies a single objective for the repeated updates; the exact population equivalence is a theoretical connection to FCP, not a claim that feedback conditioning or EM is invented here. The substantive theorems characterize feedback matching, behavioral identification, and task-improvement limitations. We do not claim novelty for the KL divergence, EM optimization, or learning from language feedback in general.

Scope. This document supplies the formulation, exact-population results, an operational feedback construction, and proofs. It is deliberately separate from the earlier proposed finite-interaction GFT-versus-RLOO separation; that claim requires its own grounding and oracle-access audit. We present one practical batch implementation, a finite-tabular empirical-MLE behavioral-convergence theorem, and a separate conditional stationary-point theorem for constrained linear-softmax batch fitting; convergence of the unrestricted neural implementation is not established. An optional

online-SGD extension gives a conditional stationary-point result in the appendix. All original exact-population assumptions and results are retained.

2 Sequential problem and feedback

2.1 Requests, executions, and optimality

Let $D \sim q^\star$ be a request. For a fixed D, an agent interacts with a finite-horizon, deterministic controlled environment with fixed initial state, horizon H, and finite action and observation sets. An execution $X = (o_1, a_1, \ldots, o_H, a_H)$ includes the complete observable history. Write $\mathcal{X}(D)$ for feasible executions. The deployed request-conditioned, potentially history-dependent policy $\pi_t(a_h \mid \mathcal{H}_h, D)$ induces the *complete-execution distribution* $p_t(x \mid D)$ on $\mathcal{X}(D)$, where $\mathcal{H}_h$ is the observed history before decision h. Thus p_t is not a separately trained model: $X \sim p_t(\cdot \mid D)$ means rolling out π_t in the environment under request D. With deterministic dynamics and a fixed initial state, $p_t(x \mid D) = \prod_{h=1}^{H} \pi_t(a_h \mid \mathcal{H}_h, D)$ for any feasible execution x.

Each request has a nonempty set $\mathcal{C}(D) \subseteq \mathcal{X}(D)$ of optimal executions. Multiple correct execution orders, tool-use sequences, and equivalent plans may be optimal. We seek

$$p_t(\mathcal{C}(D) \mid D) \longrightarrow 1, \tag{1}$$

not convergence to a user-specified distribution over correct trajectories. For finite spaces, this is equivalent to approaching the set $\mathcal{P}^\star(D) = \{p \in \Delta(\mathcal{X}(D)) : p(\mathcal{C}(D)) = 1\}$ in total variation. For a binary success objective, it also means optimal expected reward.

Why deterministic dynamics appear here. We require the posterior distribution over whole trajectories, conditional on feedback, to be realizable by a policy acting in the original environment. With independent uncontrollable transition noise, conditioning on hindsight success generally alters the law of exogenous events and is not executable by the agent. The exact posterior-reproduction claim therefore does not hold in arbitrary stochastic MDPs without additional structure.

2.2 Feedback channel and ideal-feedback generator

After execution x, a teacher or evaluator draws a feedback observation $f \in \mathcal{F}(D)$ using a channel

$$f \sim F_D(\cdot \mid x). \tag{2}$$

We assume a finite feedback alphabet for the proved results. An observation f can be a raw-language message, a scalar or categorical rating, a structured report, or a discrete encoding of an image, audio clip, or annotated past execution. The population formulation supplies no semantic parser, common embedding, or correspondence between parts of f and actions. It does not require action labels, rewards, corrections, or demonstrations, though the available feedback may contain them. Language critiques can be stochastic, abbreviated, reordered, or paraphrased. If feedback contains a complete past execution, conditioning can merely reconstruct that execution; see Section 10.3 for the resulting ideal-feedback supervision cost. The finite-space results do not automatically extend to arbitrary continuous observations.

For any execution distribution p, define its feedback marginal

$$m_p(f \mid D) := \sum_{x \in \mathcal{X}(D)} p(x \mid D) F_D(f \mid x). \tag{3}$$

We write $m_t = m_{p_t}$ for a fixed request. The second *training-time* resource, required only for requests on which this training procedure is run, is a sampler

$$f^\star \sim G^\star_D, \tag{4}$$

available on training requests. Depending on its format, sampling may or may not require knowledge of an optimal execution; a sampler containing a fully annotated optimal trajectory effectively supplies an expert demonstration. It is an assumption about supervision available for distillation, not something the algorithm has proved how to learn. It may be stochastic and need not output one canonical "success" sentence.

Definition 2.1 (Calibrated ideal feedback)**.** *The ideal-feedback generator is calibrated if, for every supported request D, there exists an optimal witness distribution $\bar{p}_D \in \mathcal{P}^\star(D)$ such that*

$$G^\star_D(f) = \sum_x \bar{p}_D(x) F_D(f \mid x) = m_{\bar{p}_D}(f \mid D). \tag{5}$$

The witness need not be observed, unique, or reproduced by the learned policy. Calibration does *not* forbid $G^\star_D$ from containing an optimal execution or a corrected action sequence; however, if such information appears in the sampled ideal feedback, access to the generator may amount to access to expert demonstrations. Conversely, in the partial-audit construction below, calibrated feedback can be sampled without an optimal execution. In either case, the generator is additional *training-time* supervision, not an input to the deployed policy.

Definition 2.2 (Coverage and exact population realizability)**.** *For every relevant D, assume (i) $\mathrm{supp}(\bar{p}_D) \subseteq \mathrm{supp}(p_0(\cdot \mid D))$; (ii) each feedback message with $G^\star_D(f) > 0$ is covered by the induced training joint; and (iii) the feedback-conditioned and clean policy classes, trained by exact population conditional maximum likelihood, can realize the corresponding full-trajectory conditional distributions in the unchanged deterministic environment.*

Part (ii) follows from (i) and calibration at initialization, and is propagated by the exact population operator below; it is stated to highlight the operational requirement. Finite-batch MLE and restricted linear or neural policy classes are not covered by this assumption.

3 Learning as feedback-distribution matching

We take the desired feedback distribution as the *policy-level training target*. A policy π induces an execution law $p_\pi(x \mid D)$ and hence the feedback law $m_\pi(f \mid D) = \sum_x p_\pi(x \mid D) F_D(f \mid x)$ from (3). Our objective is

$$\boxed{\mathcal{L}(\pi) := \mathbb{E}_{D \sim q^\star} D_{\mathrm{KL}}(G^\star_D \,\|\, m_\pi(\cdot \mid D))\,.} \tag{6}$$

The minimization is over *request-only policies*, not over a language parser or a scalarized critique. In words, we seek executions that, when evaluated by the same feedback channel as training executions, produce the *feedback distribution associated with desirable behavior*. The direction of KL matters: where the target is supported by the policy-induced channel,

$$\arg\min_\pi \mathcal{L}(\pi) = \arg\max_\pi \mathbb{E}_{D \sim q^\star,\, f \sim G^\star_D} \log \sum_x p_\pi(x \mid D) F_D(f \mid x). \tag{7}$$

Thus (6) is a latent-variable marginal-likelihood objective with the executed trajectory as latent variable and feedback as the observation. Even for free-form language feedback, no language

embedding, phrase–action correspondence, interpretation of communicative intention, or numerical reward is assumed by the population formulation. The expectation over $q^\star$ is a *training-request* objective; it is not a claim of generalization to unseen requests.

Proposition 3.1 (Feedback-relabeling invariance)**.** *Fix a request D and a bijection $h : \mathcal{F}(D) \to \widetilde{\mathcal{F}}(D)$. Relabel the actual feedback channel and desired-feedback distribution consistently: $\widetilde{F}_D(h(f) \mid x) = F_D(f \mid x)$ and $\widetilde{G}^\star_D(h(f)) = G^\star_D(f)$. For every execution law p, the feedback-KL objective and the exact population GFT update are unchanged, up to relabeling feedback values.*

Proof. The transformed marginal satisfies $\widetilde{m}_p(h(f) \mid D) = m_p(f \mid D)$, so $D_{\mathrm{KL}}(\widetilde{G}^\star_D \| \widetilde{m}_p) = D_{\mathrm{KL}}(G^\star_D \| m_p)$. On positive feedback support the Bayes posteriors satisfy $\widetilde{r}_p(x \mid D, h(f)) = r_p(x \mid D, f)$. Therefore, $\sum_{\widetilde{f}} \widetilde{G}^\star_D(\widetilde{f}) \widetilde{r}_p(x \mid D, \widetilde{f}) = \sum_f G^\star_D(f) r_p(x \mid D, f)$. □

This is a precise statement of *representation agnosticism of the population operator*, not a claim that finite-data neural models behave identically on meaningful words and arbitrary codes. The desired distribution $G^\star_D$ remains substantive supervision; calibration and feedback separation are still required for behavioral convergence.

Why an ideal-feedback distribution? The available teacher can annotate imperfect executions, but matching the distribution of their *actual* feedback would only reproduce the current policy. $G^\star_D$ supplies the target distribution instead; calibration (5) links that target to optimal behavior for the convergence theorems. A desired message that merely sounds positive need not be calibrated, and neither posterior conditioning nor feedback-KL descent alone ensures increasing task success. If ideal feedback includes a complete optimal execution, the supervision is demonstration-strength; Section 10.3 makes this explicit.

From objective to algorithm. Regard $p_t(x \mid D)$ as mixture weights and $F_D(f \mid x)$ as emission likelihoods. The EM E-step infers $r_t(x \mid D, f) \propto p_t(x \mid D) F_D(f \mid x)$, while the M-step replaces the mixture weights by $\mathbb{E}_{f \sim G^\star_D} r_t(x \mid D, f)$. The next section realizes these two steps using hindsight-conditioned imitation and request-only distillation. This *EM interpretation* is classical [1]; the research question is what it establishes about policy behavior under ambiguous, task-dependent language feedback.

4 Population GFT: conditional imitation and distillation

The algorithm is derived from the feedback-distribution objective (6). The exact population procedure has two equivalent descriptions: conditional imitation followed by distillation, and the induced request-wise distributional operator in Algorithm 1. The algorithm displays the *operator analyzed by the convergence theorems*; it is not finite-sample training pseudocode. We fix a supported request D in the algorithm because the exact update and its convergence hold separately for each request. For shared policies trained with population expectations over $D \sim q^\star$, exact conditional maximum likelihood identifies the required request-wise conditionals $q^\star$-almost everywhere under shared-policy realizability. It need not identify them on $q^\star$-null requests. When $q^\star$ is discrete, this conclusion holds at each request assigned positive probability; the fixed-request analysis below applies to any D satisfying its assumptions.

Stage I: feedback-conditioned behavior cloning. The on-policy population joint law of requests, executions, and hindsight feedback is defined by

$$D \sim q^\star, \qquad X \sim p_t(\cdot \mid D), \qquad F \sim F_D(\cdot \mid X). \tag{8}$$

Fit an auxiliary policy $\pi_t^{\mathrm{fb}}(a_h \mid \mathcal{H}_h, D, F)$ to the executed actions using conditional maximum likelihood,

$$\max_{\pi^{\mathrm{fb}}} \ \mathbb{E}_{D,X,F} \sum_{h=1}^{H} \log \pi^{\mathrm{fb}}(a_h \mid \mathcal{H}_h, D, F). \tag{9}$$

The original request is not replaced by feedback. In particular, "the permission was too broad" need not itself be an executable instruction.

Stage II: ideal-feedback sampling and distillation. The ideal-feedback-conditioned population law is induced by $D \sim q^\star$, $F^\star \sim G_D^\star$, and the Stage-I conditional execution distribution $X^+ \sim \pi_t^{\mathrm{fb}}(\cdot \mid D, F^\star)$. Exact distillation maximizes

$$\max_{\pi} \ \mathbb{E}_{D,F^\star,X^+} \sum_{h=1}^{H} \log \pi(a_h \mid \mathcal{H}_h, D). \tag{10}$$

Only $\pi_{t+1}(a_h \mid \mathcal{H}_h, D)$ acts at the next interaction round and at deployment. It receives no ideal or hindsight feedback at test time.

Algorithm 1 Population GFT: exact EM-style request-wise update

Input: Fixed supported request D; initial execution law $p_0(\cdot \mid D)$ induced by π_0; feedback channel F_D (population object); ideal-feedback law $G_D^\star$; rounds T

1: **for** $t = 0, \ldots, T-1$ **do**

2: **Feedback marginal:** For every $f \in \mathcal{F}(D)$, compute
$m_t(f \mid D) \leftarrow \sum_{x \in \mathcal{X}(D)} p_t(x \mid D) F_D(f \mid x)$.

3: **Conditional execution law:** For each f with $G_D^\star(f) > 0$, set
$r_t(x \mid D, f) \leftarrow \dfrac{p_t(x \mid D) F_D(f \mid x)}{m_t(f \mid D)} \quad (x \in \mathcal{X}(D))$.

4: **Ideal-feedback update:** For every $x \in \mathcal{X}(D)$, set
$p_{t+1}(x \mid D) \leftarrow \sum_{f : G_D^\star(f) > 0} G_D^\star(f)\, r_t(x \mid D, f)$.

5: **end for**

Output: $p_T(\cdot \mid D)$, induced by a realizable request-only policy π_T under exact population distillation

Algorithm 1 is a closed-form description of the exact population operator, not a requirement that the learner know or evaluate the feedback kernel F_D, nor a recipe involving a single training triple or a finite number of gradient updates. Its loops over x and f specify a mathematical mapping rather than operations counted toward a finite-interaction complexity bound. The learner's Stage-I access to F_D is through hindsight feedback generated after executions; Stage-II uses samples from $G_D^\star$. Calibration and initial coverage guarantee $m_t(f \mid D) > 0$ wherever $G_D^\star(f) > 0$. Under Definition 2.2, its conditional execution law r_t is exactly the Stage-I population-MLE optimum of (9); its updated execution law p_{t+1} is exactly the Stage-II population-MLE optimum of (10). Thus the request-only π_T in the output is the policy realizing p_T, not an extra model maintained alongside it. No corrected trajectory, scalar reward, or cross-modal action-alignment label is required by the general formulation; desired feedback may itself contain demonstrations, which must be accounted for. Finite-batch GFT, described in Section 8, approximates the same two stages with sampled executions and neural conditional MLE. An optional incremental-SGD variant is retained in Appendix D.

5 Exact population dynamics

As in Algorithm 1, fix a supported request D and suppress it in the notation. The equivalent conditional-imitation and distillation population objectives in (9)–(10) may average over $q^\star$ when fitting shared policies. Under exact conditional realizability, a shared-policy population MLE induces this distributional update for $q^\star$-almost every D; for discrete $q^\star$, this includes each positive-mass request. The fixed-D operator and the following theorems also hold pointwise for any request satisfying their assumptions, irrespective of the joint-training interpretation, and their bounds may be averaged under $q^\star$.

Lemma 5.1 (Posterior and multiplicative update)**.** *Under Definition 2.2, the population optimum of* (9) *realizes*

$$r_t(x \mid f) = \frac{p_t(x)F(f \mid x)}{m_t(f)} \qquad (m_t(f) > 0). \tag{11}$$

Exact distillation (10) *yields*

$$\begin{aligned} p_{t+1}(x) &= \sum_{f:G^\star(f)>0} G^\star(f) r_t(x \mid f) = p_t(x) W_t(x), \\ W_t(x) &:= \sum_{f:G^\star(f)>0} F(f \mid x) \frac{G^\star(f)}{m_t(f)}. \end{aligned} \tag{12}$$

The right-hand side is normalized and preserves support of every calibrated optimal witness $\bar{p}$.

Proof. The Stage-I joint law is $p_t(x)F(f \mid x)$, hence its exact conditional MLE is the Bayes posterior in (11). Because deterministic transition dynamics allow arbitrary feasible history-dependent action distributions to be implemented, the exact conditional policy realizes this posterior. Stage-II conditional sampling and exact cloning marginalize over $G^\star$, giving (12). Summing over x gives $\sum_x p_t(x)W_t(x) = \sum_f G^\star(f) = 1$. If $\bar{p}(x) > 0$, then every f with $F(f \mid x) > 0$ has $G^\star(f) > 0$ by $G^\star = F\bar{p}$. Such an f also has $m_t(f) > 0$ whenever $p_t(x) > 0$. Consequently $W_t(x) > 0$, and the witness support is preserved inductively. □

Theorem 5.2 (Population feedback convergence)**.** *Assume calibrated ideal feedback, coverage, and exact population conditional realizability. Let*

$$\Phi_t := D_{\mathrm{KL}}(G^\star \| m_t), \qquad V_t := D_{\mathrm{KL}}(\bar{p} \| p_t). \tag{13}$$

Then, for all $t \geq 0$,

$$\Phi_t - \Phi_{t+1} \geq D_{\mathrm{KL}}(p_{t+1} \| p_t) \geq 0, \qquad V_t - V_{t+1} \geq \Phi_t. \tag{14}$$

Consequently Φ_t is nonincreasing, and

$$D_{\mathrm{KL}}(G^\star \| m_T) \leq \frac{D_{\mathrm{KL}}(\bar{p} \| p_0)}{T+1} \qquad (T \geq 0). \tag{15}$$

Proof. The first inequality is the EM-type monotonicity argument. Writing $r_t(x \mid f)$ for the Bayes posterior, observe that

$$\frac{m_{t+1}(f)}{m_t(f)} = \sum_x r_t(x \mid f) W_t(x).$$

Jensen's inequality and then a change of order give

$$\begin{aligned}\Phi_t - \Phi_{t+1} &= \sum_f G^\star(f) \log \frac{m_{t+1}(f)}{m_t(f)} \\ &\geq \sum_{f,x} G^\star(f) r_t(x \mid f) \log W_t(x) \\ &= \sum_x p_{t+1}(x) \log \frac{p_{t+1}(x)}{p_t(x)}.\end{aligned}$$

For the second inequality, (12) and Jensen give

$$\begin{aligned}V_t - V_{t+1} &= \sum_x \bar{p}(x) \log W_t(x) \\ &\geq \sum_{x,f} \bar{p}(x) F(f \mid x) \log \frac{G^\star(f)}{m_t(f)} \\ &= \sum_f G^\star(f) \log \frac{G^\star(f)}{m_t(f)} = \Phi_t.\end{aligned}$$

All logarithms here are evaluated on the positive support of the corresponding distribution. Summing the second inequality over $t = 0, \ldots, T$ gives $\sum_{t=0}^{T} \Phi_t \leq V_0$; by monotonicity $(T+1)\Phi_T \leq \sum_{t=0}^{T} \Phi_t$. This proves (15). □

EM interpretation and scope of novelty. Treat x as a latent mixture component, $F(f \mid x)$ as its emission likelihood, and $G^\star(f)$ as the target distribution of observed feedback. The posterior $r_t(x \mid f)$ is the EM E-step, while $p_{t+1}(x) = \sum_f G^\star(f) r_t(x \mid f)$ is the exact mixture-weight M-step. Therefore the monotonicity mechanism is a standard EM-type likelihood argument [1]; the substantive behavioral guarantee additionally requires calibration and optimality-separating feedback.

What converges? We have proved convergence of the feedback marginal, not of the parameters of a particular neural policy or toward a preselected trajectory mixture. By Pinsker,

$$\mathrm{TV}(G^\star, m_T) \leq \sqrt{\frac{V_0}{2(T+1)}}. \tag{16}$$

Matching feedback does not, by itself, force correct behavior. *Nor does monotone decrease of Φ_t imply monotone increase of $p_t(\mathcal{C})$; the latter can temporarily decrease even when the sufficient conditions for eventual optimality hold.*

6 When does feedback matching imply optimality?

6.1 A sufficient statistical separation condition

Let $\mathcal{C} \subset \mathcal{X}$ be the optimal executions for the fixed request. The weakest identification property relevant to Theorem 5.2 is

$$m_p = G^\star \implies p(\mathcal{C}) = 1 \qquad \text{for every execution distribution } p \text{ on accessible support.} \tag{17}$$

It is insufficient to require merely $F(\cdot \mid x) \neq G^\star$ for each individual bad x: mixtures of bad executions may still reproduce $G^\star$.

Definition 6.1 (Optimality-separating feedback statistic)**.** *A feedback channel is γ-optimality separating for the request if there exists a statistic $s : \mathcal{F} \to [0,1]$, a constant $a \in [0,1]$, and $\gamma > 0$ such that*

$$\mathbb{E}[s(F) \mid X = x] = a \quad \forall x \in \mathcal{C}, \qquad \mathbb{E}[s(F) \mid X = x] \le a - \gamma \quad \forall x \notin \mathcal{C}. \tag{18}$$

The statistic is a certificate used by the analysis: the GFT algorithm is neither given s nor trained to predict it.

The shared optimal mean in (18) is important. Without it, mixtures of high-scoring optimal and low-scoring incorrect executions could cancel a feedback gap. The condition allows overlap between the language distributions of individual correct and incorrect executions.

Lemma 6.2 (Separation of every suboptimal mixture)**.** *If $G^\star$ is calibrated and* (18) *holds, then every execution distribution p satisfies*

$$\gamma p(\mathcal{C}^c) \le \mathbb{E}_{G^\star} s - \mathbb{E}_{m_p} s \le \mathrm{TV}(G^\star, m_p). \tag{19}$$

In particular (17) *holds.*

Proof. Calibration gives $\mathbb{E}_{G^\star} s = \mathbb{E}_{\bar{p}} \mathbb{E}[s \mid X] = a$. Averaging (18) under p gives $\mathbb{E}_{m_p} s \le a - \gamma p(\mathcal{C}^c)$. For a statistic valued in $[0,1]$, its expectation difference under two laws is bounded by their total variation distance. □

Theorem 6.3 (Convergence to optimal behavior)**.** *Under the assumptions of Theorem 5.2 and the γ-optimality separating condition* (18)*, for every $T \ge 0$,*

$$p_T(\mathcal{C}^c) \le \min\left\{1, \frac{1}{\gamma}\sqrt{\frac{D_{\mathrm{KL}}(\bar{p} \| p_0)}{2(T+1)}}\right\}. \tag{20}$$

Hence $p_T(\mathcal{C}) \to 1$. For binary reward $R(x) = \mathbf{1}\{x \in \mathcal{C}\}$, the expected reward tends to its optimal value 1*. The theorem does not imply convergence to any particular optimal mixture $p^\star$.*

Proof. Combine (19) with (16). □

6.2 When is task success monotone?

Calibration alone does not imply success improvement in one iteration, as the example in Section 6.3 shows. A stronger condition does suffice, and it holds for our truthful partial-audit construction.

Proposition 6.4 (Monotone success under common optimal feedback)**.** *Under the exact population assumptions of Theorem 5.2, suppose every optimal execution induces the* same *feedback law:*

$$F(\cdot \mid x) = G^\star \qquad \text{for all } x \in \mathcal{C}. \tag{21}$$

Then $p_{t+1}(\mathcal{C}) \ge p_t(\mathcal{C})$ for every t. Under optimality separation, these nondecreasing success probabilities converge to one. Condition (21) *is satisfied by the truthful partial-audit channel in Section 7, but is not implied by calibration or by the γ-separation condition alone.*

Proof. Writing $q_t = p_t(\mathcal{C})$ and using (12) and (21),

$$q_{t+1} = q_t \sum_{f : G^\star(f) > 0} \frac{G^\star(f)^2}{m_t(f)} \ge q_t \frac{\left(\sum_f G^\star(f)\right)^2}{\sum_{f : G^\star(f) > 0} m_t(f)} \ge q_t,$$

by Cauchy–Schwarz and $\sum_{f:G^\star(f)>0} m_t(f) \le 1$. Calibration and initial coverage ensure every denominator shown is positive. Theorem 6.3 gives eventual optimality when separation holds. In the partial-audit channel, every optimal execution satisfies all requirements, the entire audit-subset law depends only on the request, and the wording kernel depends only on the selected truth values; consequently all optimal executions induce the same feedback law. □

6.3 Feedback-KL descent need not improve task success

A counterexample even with $\gamma > 0$. Consider three feasible executions x_A, x_B, x_C, of which x_A, x_B are optimal and x_C is suboptimal. The feedback alphabet is $\{f_A, f_B, f_0\}$, and the feedback channel, with columns in that order, is

	f_A	f_B	f_0	$\mathbb{E}[s(F) \mid x]$
x_A (optimal)	0.8	0	0.2	0.8
x_B (optimal)	0	0.8	0.2	0.8
x_C (suboptimal)	0	0.7	0.3	0.7

Here $s(f_A) = s(f_B) = 1$ and $s(f_0) = 0$. Thus Definition 6.1 holds with $a = 0.8$ and $\gamma = 0.1$. Take $G^\star = F(\cdot \mid x_B)$ and $p_0(x_A, x_B, x_C) = (0.98, 0.001, 0.019)$. Calibration, initial coverage, and exact realizability all hold in a deterministic one-decision environment. Applying the exact update (12) yields

$$p_0(\mathcal{C}) = 0.981, \qquad p_1 \approx (0.19416,\ 0.04559,\ 0.76026),$$
$$p_1(\mathcal{C}) \approx 0.23974, \quad \Phi_0 \approx 3.22886, \quad \Phi_1 \approx 0.20864.$$

Success therefore falls from 98.1% to about 24.0% while feedback KL decreases. Theorem 6.3 nevertheless ensures $p_t(\mathcal{C}) \to 1$ as $t \to \infty$. This example establishes that *monotone feedback matching and asymptotic optimality do not require, or imply, monotone policy improvement.* The numerical values follow directly from (12), not from a simulation or a different learning algorithm.

6.4 What if the optimality gap becomes arbitrarily small?

Uniform $\gamma > 0$ can fail in continuous-quality tasks. Let $R : \mathcal{X} \to [0, 1]$, $R^\star = \max_x R(x)$, and fix $\varepsilon > 0$. A weaker requirement is that there exist $s \in [0, 1]$, a, and $\gamma_\varepsilon > 0$ such that $\mathbb{E}[s \mid x] \le a$ for every execution, with equality on every optimal execution, and $\mathbb{E}[s \mid x] \le a - \gamma_\varepsilon$ whenever $R(x) \le R^\star - \varepsilon$. Repeating the argument gives

$$\Pr_{p_T}\{R(X) \le R^\star - \varepsilon\} \le \frac{1}{\gamma_\varepsilon}\sqrt{\frac{V_0}{2(T+1)}}. \tag{22}$$

This proves asymptotic concentration on ε-optimal outcomes for each fixed ε; it does not require a uniform gap over trajectories with arbitrarily small reward suboptimality.

7 An illustrative language channel: truthful partial requirement reports

The optimality-separating condition need not be asserted abstractly. We now construct an interpretable feedback process that satisfies it, without requiring the teacher to diagnose every requirement on every execution.

7.1 Task-relative requirements and a partial auditor

For each request D, define $m = m(D) \geq 1$ verifiable requirements, with indicators

$$r_j(D, x) \in \{0, 1\}, \quad j = 1, \ldots, m, \qquad \mathcal{C}(D) = \{x : r_j(D, x) = 1 \ \forall j\}. \tag{23}$$

For an email-and-permissions task, requirements might include the recipient, attachment version, and permission level. Different trajectories may satisfy all requirements through different tool sequences.

After execution, the teacher selects an audit subset $S \subseteq \{1, \ldots, m\}$ from a distribution Q_D that depends on the request, *not* on the execution or its achieved truth values. Specifically,

$$\Pr(S = s \mid D, x) = Q_D(s), \qquad \kappa_j(D) := \Pr_{S \sim Q_D}(j \in S) \geq \kappa_{\min}(D) > 0. \tag{24}$$

The full-law independence in (24) is needed to calibrate a request-only ideal-feedback sampler for *every* optimal execution. Merely requiring execution-independent marginal inclusion rates $\kappa_j(D)$ would suffice for the score-gap calculation below, but not for that calibration claim. There is no requirement that inclusion events for different j be independent. The teacher then truthfully reports the satisfaction or violation of every $j \in S$, in arbitrary order and arbitrary semantically faithful language. Conditional on D, S and these truth values, the distribution of report wording and ordering does not depend on any other property of the execution. It may omit the other requirements entirely. Feedback chunks may be paraphrased, grouped, or shuffled; no action–chunk alignment is provided.

Example. Request: "Send the Q3 summary to Alice and give Bob read access." An execution sends Q3 to Alice but gives Bob edit rights. Depending on S, feedback might say "The document and recipient were correct," or "Bob received edit permission rather than read-only access." Both are accurate partial reports of the same execution.

An ideal-feedback sampler can be implemented at the level of the request: sample S by the same audit-selection procedure and verbalize the selected requirements as satisfied. When every requirement is achievable, this samples the feedback distribution of any optimal execution, without needing to construct that execution. Sampling ideal feedback still assumes access to accurate descriptions of satisfied requirements; it is not free information.

Proposition 7.1 (Truthful partial audits separate optimal behavior)**.** *Under* (23)–(24)*, suppose the language explicitly and correctly conveys the polarity of each reported requirement (though not its action alignment). Define*

$$s_D(F) := \frac{\#\{j \in S : \textit{requirement } j \textit{ is reported satisfied}\}}{m}. \tag{25}$$

Then $0 \leq s_D \leq 1$ *and*

$$\mathbb{E}[s_D(F) \mid D, X = x] = \frac{1}{m} \sum_{j=1}^{m} \kappa_j(D) r_j(D, x). \tag{26}$$

Consequently the channel satisfies Definition 6.1 with

$$a_D = \frac{1}{m} \sum_{j=1}^{m} \kappa_j(D), \qquad \gamma_D = \frac{\kappa_{\min}(D)}{m(D)}. \tag{27}$$

The ideal-feedback generator described above is calibrated for every optimal witness distribution.

Proof. By truthful reporting, the numerator in (25) is $\sum_j \mathbf{1}\{j \in S\} r_j(D, x)$. Taking expectations yields (26); independence across requirements is unnecessary. Every optimal execution has all $r_j = 1$ and therefore expectation a_D. Every suboptimal execution fails at least one requirement, causing a loss of at least $\kappa_{\min}/m$. For any optimal execution, every sampled requirement is described as satisfied. Since the *entire* audit-subset law is Q_D for every execution, and the wording kernel depends only on D, S, and the selected truth values, ideal-feedback sampling from the request reproduces exactly the teacher's conditional law for every optimal execution. This last conclusion would not follow from the marginal rates κ_j alone. □

Corollary 7.2 (Explicit optimality rate under truthful partial audits)**.** *Under the exact population assumptions and the partial-audit channel,*

$$p_T(\mathcal{C}(D)^c \mid D) \le \frac{m(D)}{\kappa_{\min}(D)} \sqrt{\frac{D_{\mathrm{KL}}(\bar{p}_D \| p_0(\cdot \mid D))}{2(T+1)}}. \tag{28}$$

The bound is understood as capped at one.

A less pessimistic graded-quality rate. Suppose all requirements have the same inclusion probability $\kappa > 0$ and quality is measured by the fraction satisfied,

$$R_D(x) := \frac{1}{m} \sum_j r_j(D, x), \qquad R_D^\star = 1. \tag{29}$$

Then (26) gives $\mathbb{E}[s_D \mid x] = \kappa R_D(x)$ and $\mathbb{E}_{G_D^\star} s_D = \kappa$. Therefore,

$$1 - \mathbb{E}_{p_T} R_D(X) \le \frac{1}{\kappa} \sqrt{\frac{D_{\mathrm{KL}}(\bar{p}_D \| p_0(\cdot \mid D))}{2(T+1)}}. \tag{30}$$

This quality bound has no explicit factor of m; exact all-requirements success is a stricter target than high average requirement satisfaction.

7.2 Why incomplete truthful feedback is sufficient

In this language-audit example, the teacher may omit the one failed requirement in a particular episode. Thus a suboptimal execution can receive feedback that is indistinguishable from a piece of ideal feedback, despite zero teacher misreporting. The proof does not require identifying optimality from a single message. It requires that any persistently violated requirement has a positive probability of being reported, producing a detectable difference in the feedback distribution. All calculations concern complete raw-language messages; they do not reveal segment–action alignment to the learner. The existence of s_D is a property of a semantically faithful reporting mechanism, not a supplied semantic parser.

8 Practical GFT: sampled batch rounds

The population operator is a mathematical idealization: it assumes exact conditional fitting and exact distillation. A practical implementation alternates supervised *batch* fits, using separate models for feedback-conditioned generation and request-only execution. Internal minibatch SGD is allowed; "batch" here means that each outer round collects experience and fits the two supervised objectives

before collecting the next on-policy round. We choose *one* practical batch procedure for the main paper, rather than treating every update schedule as a separate method.

Algorithm 2 Batch GFT: finite-data conditional learning and distillation

Input: Training requests $q^\star$; actual feedback oracle F_D; training-time desired feedback sampler $G^\star_D$; clean policy π_{θ_0}; auxiliary policy π^{fb}_ϕ; batch sizes N_t, M_t; rounds T

1: **for** $t = 0, \dots, T-1$ **do**

2: **On-policy data:** Sample N_t training requests D_i, roll out $X_i \sim p_{\theta_t}(\cdot \mid D_i)$, and obtain *actual* hindsight feedback $f_i \sim F_{D_i}(\cdot \mid X_i)$.

3: **Conditional fit:** Update ϕ by supervised minibatch optimization of $-\sum_{i=1}^{N_t}\sum_{h=1}^{H} \log \pi^{\mathrm{fb}}_\phi(a_{i,h} \mid \mathcal{H}_{i,h}, D_i, f_i)$, producing ϕ_t.

4: **Desired-feedback rollouts:** Independently sample M_t training requests D'_j and $f^\star_j \sim G^\star_{D'_j}$; roll out $X^+_j \sim p^{\mathrm{fb}}_{\phi_t}(\cdot \mid D'_j, f^\star_j)$.

5: **Distillation:** Update θ by supervised minibatch optimization of $-\sum_{j=1}^{M_t}\sum_{h=1}^{H} \log \pi_\theta(a^+_{j,h} \mid \mathcal{H}^+_{j,h}, D'_j)$, producing θ_{t+1}.

6: **end for**

Output: Request-only policy π_{θ_T} for deployment

The desired feedback $f^\star_j$ in the third step is *not* silently substituted for actual feedback f_i in the conditional fit. Both execution batches take place in the same decision environment; the conditional policy changes the conditioning information, not the tools or transition dynamics. Stage-I fitting and request-only distillation are approximations to (9)–(10). The architecture, number of SGD steps, optional warm starts, and batch schedule are implementation choices, not established to be equivalent to exact population EM. At inference only $\pi_{\theta_T}(a_h \mid \mathcal{H}_h, D)$ is used.

Guarantees for the same sampled outer loop. Algorithm 2 is a training recipe, not a convergence theorem for arbitrary function approximators. The two results below cover different, explicitly restricted specializations:

Model and fit	Additional conditions	Proven conclusion
Unrestricted finite tables	Exact empirical frequency fits; sufficient growing actual and synthetic batches; calibrated coverage	High-probability feedback-KL and behavioral rates under separation
Fixed-feature linear softmax	Constrained approximate global empirical fits; realizable posteriors; coverage and strong M-step curvature	High-probability approach to constrained stationary points of feedback KL
Generic neural networks	No corresponding assumptions established	No convergence theorem claimed

8.1 Finite-tabular batch convergence: a constructive baseline

The exact-operator theorem also has a finite-sample analogue for an *unrestricted finite-tabular specialization* of Algorithm 2. This result does not assume that a neural conditional model can be trained to a specified error. Instead, both conditional fitting and distillation are *exact empirical maximum-likelihood fits* (equivalently, empirical conditional and marginal trajectory frequencies). It controls their statistical errors directly. All statements in this subsection fix one request D; write $n = |\mathcal{X}(D)|$ and $k = |\mathcal{F}(D)|$.

Theorem 8.1 (High-probability convergence of finite-tabular Batch GFT)**.** *Assume the calibration and initial coverage hypotheses of Theorem 5.2, with finite execution and feedback sets and a*

fixed supported request. Let $S = \operatorname{supp}(\bar{p})$, $v = D_{\mathrm{KL}}(\bar{p}\|p_0)$, $h = -\sum_{x\in S}\bar{p}(x)\log\bar{p}(x)$, *and* $a = \min_{x\in S}\bar{p}(x) > 0$. *Define the positive (possibly extremely small) constants*

$$b := \exp\left(-\frac{v+1+h}{a}\right), \qquad \beta := \frac{ba}{2k}, \qquad e_t := \frac{\beta}{4(t+1)^3}. \tag{31}$$

Fix a failure probability $\delta \in (0,1)$. *At round* t, *collect* N_t *fresh independent actual execution–feedback pairs from the current policy, fit their empirical conditional law* $\widehat{r}_t(x \mid f)$ *(arbitrary at unseen* f*), sample* M_t *independent executions from* $\sum_f G^\star(f)\widehat{r}_t(\cdot \mid f)$, *and exactly fit the clean empirical execution law* p_{t+1}. *Suppose*

$$N_t \geq \frac{8}{b^2 e_t^2}\left[(nk+1)\log 2 + \log\frac{2(t+1)(t+2)}{\delta}\right], \tag{32}$$

$$M_t \geq \frac{2}{e_t^2}\left[(n+1)\log 2 + \log\frac{2(t+1)(t+2)}{\delta}\right]. \tag{33}$$

Then, with probability at least $1-\delta$, *simultaneously for every* $T \geq 0$,

$$D_{\mathrm{KL}}(G^\star\|m_{p_T}) \leq \frac{v+7/4}{T+1}. \tag{34}$$

If the γ*-optimality-separation assumption of Theorem 6.3 also holds, on the same event*

$$p_T(\mathcal{C}^c) \leq \min\left\{1, \frac{1}{\gamma}\sqrt{\frac{v+7/4}{2(T+1)}}\right\}. \tag{35}$$

Proof sketch. Write $q_t = \mathcal{T}(p_t)$ for the exact population update and $\widehat{q}_t = \sum_f G^\star(f)\widehat{r}_t(\cdot \mid f)$ for the fitted auxiliary mixture. Conditional empirical-distribution concentration, then concentration of pseudo-trajectories around $\widehat{q}_t$, imply $\mathrm{TV}(p_{t+1}, q_t) \leq e_t$ simultaneously over all rounds on an event of probability at least $1-\delta$. Crucially, this uses the lower bound $m_{p_t}(f) \geq bG^\star(f)$ on ideal-supported feedback, not a lower bound on all incorrect-execution probabilities. The exact update and $\bar{p}$-coverage give $q_t(x) \geq 2\beta$ for $x \in S$, provided $D_{\mathrm{KL}}(\bar{p}\|p_t) \leq v+1$. The sampling error then gives $p_{t+1}(x) \geq \beta$ on S. On these supported coordinates, log-KL potentials are Lipschitz in total variation, yielding

$$V_t - V_{t+1} \geq \Phi_t - \frac{2e_t}{\beta}, \qquad \Phi_{t+1} \leq \Phi_t + \frac{2e_t}{\beta}.$$

Since $\sum_t e_t \leq 3\beta/8$, induction maintains $V_t \leq v+1$, closing the support argument. Summing and using near-monotonicity gives

$$(T+1)\Phi_T \leq v + \frac{2}{\beta}\sum_{t=0}^{T}(t+2)e_t \leq v + \frac{7}{4}.$$

The separation bound and Pinsker's inequality give (35). Full details, including the conditional concentration argument, are in Appendix A. □

Scope and cost. This is a genuine sampled-interaction theorem for a *tabular implementation* of the batch stages, analyzed for a *fixed request*, not a uniform finite-data generalization theorem for shared policies across unseen requests or a result for unrestricted neural SGD. The schedule depends on the unknown optimal witness and initial inverse coverage through b and β; n and k can

grow exponentially with horizon and with the number of possible free-form feedback strings. The polynomial dependence on the outer iteration count t therefore *does not* imply a useful polynomial interaction bound in horizon, feedback complexity, or inverse optimal-policy coverage. Generic neural batch convergence remains unproved. The next subsection gives a separate restricted-parameter stationary-point result, not a strengthening of the global tabular theorem. The gradient identity below gives a different, local connection to practical neural training. The optional online theorem in Appendix D analyzes a different optimization schedule.

8.2 Finite-batch stationarity for linear-softmax Batch GFT

The tabular theorem above is global but assumes an unrestricted trajectory table. We next analyze a *different, restricted* specialization of the *same batch outer loop*, with a clean linear-softmax policy

$$\pi_\theta(a \mid \mathcal{H}, D) = \frac{\exp\langle \theta, \varphi(\mathcal{H}, D, a)\rangle}{\sum_b \exp\langle \theta, \varphi(\mathcal{H}, D, b)\rangle}, \qquad \theta \in \Theta, \tag{36}$$

and an analogous feedback-conditioned linear-softmax policy $\pi_\phi^{\mathrm{fb}}(a \mid \mathcal{H}, D, f)$, $\phi \in K$. The features are fixed and bounded (not learned). The sets Θ and K are compact and convex. In this theorem, the two empirical supervised objectives in Algorithm 2 are approximately *globally* optimized over these sets. This is a constrained specialization: the unconstrained pseudocode is covered only when the constraints are inactive. In this specialization, a convex supervised fitting problem is *not* evidence that its posterior is realizable, that the feedback objective is concave, or that a given SGD budget attains the prescribed fitting accuracy.

For $\theta \in \Theta$, let $r_\theta(x \mid D, f)$ denote the exact execution posterior and define the usual EM surrogate

$$Q_\theta(v) = \mathbb{E}_{D\sim q^\star,\, f\sim G_D^\star,\, x\sim r_\theta(\cdot\mid D,f)} \log p_v(x \mid D), \qquad v \in \Theta. \tag{37}$$

Write $J(\theta) = -\mathbb{E}_D D_{\mathrm{KL}}(G_D^\star \| m_\theta(\cdot \mid D))$ as in (43). At each round let $\widehat{\ell}_t(\phi)$ be the empirical conditional negative log likelihood from N_t fresh current-policy executions, and $\widehat{Q}_t(v)$ the empirical clean-policy log likelihood on M_t fresh ideal-feedback-conditioned executions from the newly fitted auxiliary policy. We make the following *additional* assumptions:

(B1) Finite bounded setting. Requests, actions, executions, and feedback are finite, with fixed deterministic finite-horizon dynamics and fixed bounded linear-softmax features. On $\Theta \times K$, full feasible-execution log likelihoods are bounded, the associated per-trajectory losses are uniformly Lipschitz in their parameters, and J is continuously differentiable.

(B2) Uniform posterior realization and desired-feedback coverage. For every $\theta \in \Theta$ a parameter $\phi^\star(\theta) \in K$ realizes the exact $r_\theta(\cdot \mid D, f)$ for every (D, f) with $q^\star(D) m_\theta(f \mid D) > 0$. Also $G_D^\star \ll m_\theta(\cdot \mid D)$ and $\kappa := \sup_{\theta, D, f: G_D^\star(f) > 0} G_D^\star(f)/m_\theta(f \mid D) < \infty$.

(B3) Uniform clean-surrogate curvature. For some $\mu > 0$, the map $v \mapsto Q_\theta(v)$ is μ-strongly concave on Θ for every $\theta \in \Theta$. Its gradient in v is uniformly L_Q-Lipschitz. This is a uniform Fisher-information/nonredundancy condition, *not* a consequence of linearity alone.

(B4) Constrained empirical fitting. The auxiliary fit has empirical NLL excess at most $a_t \geq 0$, and the student fit has empirical negative log-likelihood excess at most $b_t \geq 0$. All training and distillation samples are fresh at the specified round, conditional on the previous fitting stage.

An ordinary finite-parameter softmax has strictly positive action probabilities. Therefore, the posterior-realization assumption can fail for truthful feedback that assigns exactly zero likelihood

to an incompatible execution; such settings are *not* automatically covered. Posterior realization and uniform curvature can hold in nondegenerate positive-noise finite examples, but both are substantive restrictions. For example, in a one-step two-action problem with $p_\theta(1) = \sigma(\theta)$, let $F(+ \mid 1) = 0.9$ and $F(+ \mid 0) = 0.1$. Then the exact posterior odds are $\operatorname{logit} r_\theta(1 \mid +) = \theta + \log 9$ and $\operatorname{logit} r_\theta(1 \mid -) = \theta - \log 9$, realized by a two-feature auxiliary logistic model with a feedback indicator. On any compact clean-parameter interval, $Q_\theta(v)$ has second derivative $-\sigma(v)(1-\sigma(v))$, uniformly bounded away from zero. Thus the restricted assumptions are jointly satisfiable without an unrestricted trajectory table.

Theorem 8.2 (Finite-batch linear-softmax stationary-point convergence)**.** *Assume (B1)–(B4). Suppose at round t, conditional on the past, the following uniform empirical deviations hold:*

$$\sup_{\phi \in K} |\widehat{\ell}_t(\phi) - \ell_{\theta_t}(\phi)| \le u_t, \tag{38}$$

$$\sup_{v \in \Theta} |\widehat{Q}_t(v) - \widetilde{Q}_{\theta_t,\phi_t}(v)| \le z_t, \tag{39}$$

where ℓ_θ is the true conditional NLL under $D \sim q^\star$, $X \sim p_\theta(\cdot \mid D)$, $f \sim F_D(\cdot \mid X)$, and $\widetilde{Q}_{\theta,\phi}$ uses executions from the learned π^{fb}_ϕ *under $f \sim G^\star_D$. Let L_x bound $|\log p_v(x \mid D)|$ and put*

$$\eta_t := 4L_x \sqrt{\frac{\kappa(2u_t + a_t)}{2}} + 2z_t + b_t. \tag{40}$$

On any event where (38) *holds for every round and $\sum_t \eta_t < \infty$, the Batch GFT iterates satisfy*

$$\sum_{t=0}^{\infty} \|\theta_{t+1} - \theta_t\|^2 < \infty, \qquad \big\|\theta_t - \Pi_\Theta\big(\theta_t + \nabla J(\theta_t)\big)\big\| \longrightarrow 0. \tag{41}$$

Every accumulation point is a constrained stationary point of J. If the iterates eventually remain a fixed positive distance from $\partial\Theta$, then $\|\nabla J(\theta_t)\| \to 0$. In particular, for any $\delta \in (0,1)$, uniformly bounded linear-softmax losses and compact finite-dimensional parameter sets permit batch sizes

$$N_t = \widetilde{O}\big((t+1)^8\big), \qquad M_t = \widetilde{O}\big((t+1)^4\big) \tag{42}$$

(with constants depending on parameter dimensions, loss bounds, and δ), provided *that the two constrained convex empirical problems are also solved to optimization excesses $a_t = O((t+1)^{-4})$ and $b_t = O((t+1)^{-2})$. Under these separate statistical and optimization conditions,* (41) *holds with probability at least $1 - \delta$. Choosing batch sizes alone does not guarantee the optimization tolerances. This is stationarity, not global feedback matching or behavioral optimality.*

Proof outline. Exact posterior realizability and conditional maximum likelihood imply that the population auxiliary excess loss equals an average conditional KL. Uniform deviation and stage-I optimization bound that excess by $2u_t + a_t$. Changing its averaging law from the actual feedback marginal m_{θ_t} to $G^\star$, then applying Pinsker, bounds the ideal-feedback-average posterior total variation by $\sqrt{\kappa(2u_t + a_t)/2}$. Since $|\log p_v| \le L_x$, the resulting surrogate differs uniformly from Q_{θ_t} by at most $2L_x$ times this total variation. Adding the second-stage empirical and optimization errors shows

$$Q_{\theta_t}(\theta_{t+1}) \ge \max_{v \in \Theta} Q_{\theta_t}(v) - \eta_t.$$

The usual EM lower bound gives $J(\theta_{t+1}) - J(\theta_t) \ge Q_{\theta_t}(\theta_{t+1}) - Q_{\theta_t}(\theta_t)$. Strong concavity and approximate maximization yield

$$J(\theta_{t+1}) - J(\theta_t) \ge \frac{\mu}{4}\|\theta_{t+1} - \theta_t\|^2 - 2\eta_t.$$

Summation proves square-summable increments. Strong concavity also puts θ_{t+1} within $\sqrt{2\eta_t/\mu}$ of the maximizer of Q_{θ_t}. Combining these observations with the first-order EM identity $\nabla J(\theta_t) = \nabla_v Q_{\theta_t}(v)|_{v=\theta_t}$ proves the projected-gradient limit. Uniform covering-number/Hoeffding bounds, applied conditionally at each stage and union-bounded over rounds, give the displayed growing-batch schedule. Appendix B contains the inequalities and sample-size details. □

Scope and non-implications. This theorem analyzes the two-fit *batch* update only under fixed-feature linear-softmax, controlled convex empirical optimization, uniform posterior realizability, and strong curvature. Unlike the population behavioral theorem, it needs no optimality-separating statistic: its conclusion concerns stationarity of J, not correctness. It does not transfer the global tabular guarantee to restricted policies or prove convergence for generic neural models, fixed optimization budgets, or unconstrained iterates that escape the parameter set. It also does not apply directly to the paper's truthful partial-audit example when the auxiliary softmax cannot realize posterior zeros.

9 A gradient view of the feedback-matching objective

The exact-population theory above minimizes the mismatch between the teacher's ideal-feedback distribution and the feedback distribution induced by the clean policy. The same objective also admits a direct gradient formulation useful for diagnosing finite-sample distillation. A single fresh actor-gradient step has a clean interpretation, but arbitrarily many steps against a frozen pseudo-trajectory batch do not inherit it. This section retains the fixed feedback channel and model-based execution law of Sections 3–5. Parameter-space regularity conditions introduced below are *additional* to the original population theory.

Let π_θ be a differentiable clean policy with execution law $p_\theta(x \mid D)$ and feedback marginal $m_\theta(f \mid D)$. Write the negative of the objective (6) as

$$J(\theta) := -\mathbb{E}_{D\sim q^\star} D_{\mathrm{KL}}(G_D^\star \| m_\theta(\cdot \mid D)), \qquad s_\theta(x, D) := \nabla_\theta \log p_\theta(x \mid D). \tag{43}$$

For the following identities assume that $G_D^\star(f) > 0$ implies $m_\theta(f \mid D) > 0$, that differentiation and expectation can be interchanged, and that the execution support and parameter derivatives are regular enough for $\nabla p_\theta(x \mid D) = p_\theta(x \mid D) s_\theta(x, D)$ on the relevant support. For policies with zero-probability executions, the score identities are evaluated only on their positive support; in the online algorithm, generated auxiliary trajectories must also admit a well-defined clean-policy score (e.g., a full-support clean softmax policy). In the deterministic environment,

$$s_\theta(x, D) = \sum_{h=1}^{H} \nabla_\theta \log \pi_\theta(a_h \mid \mathcal{H}_h, D).$$

Write $r_\theta(x \mid D, f)$ for the exact conditional execution posterior in Lemma 5.1, now using p_θ.

Proposition 9.1 (Policy-gradient and first-distillation identities)**.** *For every parameter value satisfying the preceding support and differentiability conditions,*

$$\nabla_\theta J(\theta) = \mathbb{E}_{\substack{D\sim q^\star,\, f^\star \sim G_D^\star \\ x^+ \sim r_\theta(\cdot \mid D, f^\star)}} s_\theta(x^+, D), \tag{44}$$

$$= \mathbb{E}_{\substack{D\sim q^\star,\, x\sim p_\theta(\cdot \mid D) \\ f\sim F_D(\cdot \mid x)}} \left[\frac{G_D^\star(f)}{m_\theta(f \mid D)} s_\theta(x, D) \right]. \tag{45}$$

Fix θ_t, and let $p_t^+(x \mid D) := \sum_f G_D^\star(f) r_{\theta_t}(x \mid D, f)$. Holding this pseudo-trajectory distribution fixed, define the conditional-distillation surrogate

$$L_t(\theta) := \mathbb{E}_{D\sim q^\star,\, x^+\sim p_t^+(\cdot|D)} \log p_\theta(x^+ \mid D).$$

Then $\nabla L_t(\theta_t) = \nabla J(\theta_t)$. The identity does not *assert equality of these two gradients at arbitrary $\theta \neq \theta_t$.*

Proof. For each request, differentiating the forward KL objective gives

$$\begin{aligned}
\nabla J_D(\theta) &= \sum_f \frac{G_D^\star(f)}{m_\theta(f \mid D)} \nabla m_\theta(f \mid D) \\
&= \sum_{f,x} \frac{G_D^\star(f)}{m_\theta(f \mid D)} F_D(f \mid x) p_\theta(x \mid D) s_\theta(x, D) \\
&= \sum_f G_D^\star(f) \sum_x r_\theta(x \mid D, f) s_\theta(x, D).
\end{aligned}$$

The first and second expressions give (44) and (45). Since p_t^+ is treated as fixed when differentiating L_t, its score expectation evaluated at θ_t is exactly (44). □

Interpretation. At the current actor, *one* supervised distillation gradient with fresh, exact ideal-feedback-conditioned rollouts is an unbiased stochastic gradient of a fixed mathematical objective J. This is an equality of *first derivatives* at the current parameter value; it is not an equality of the full supervised objective and J. Once the actor changes, an unchanged distillation batch becomes stale; repeated steps on the same frozen surrogate need not follow ∇J.

Proposition 9.2 (Gradient bias from an approximate feedback-conditioned model)**.** *Let $r_\phi(\cdot \mid D, f)$ be the trajectory distribution induced by a learned feedback-conditioned policy. If $\|s_\theta(x, D)\| \le B$, define*

$$\delta(\theta, \phi) := \mathbb{E}_{D\sim q^\star,\, f\sim G_D^\star} \mathrm{TV}\left(r_\phi(\cdot \mid D, f), r_\theta(\cdot \mid D, f)\right).$$

The mean clean-policy score from auxiliary ideal-feedback rollouts, denoted $g(\theta, \phi)$, satisfies

$$\|g(\theta, \phi) - \nabla J(\theta)\| \le 2B\, \delta(\theta, \phi). \tag{46}$$

Proof. For any bounded vector-valued function u with $\|u\| \le B$, the difference in its expectations under two probability laws has norm at most $2B\,\mathrm{TV}(P, Q)$. Apply this with $u = s_\theta$, then average over D, f. □

10 Relation to FCP and the role of training-time privileged feedback

10.1 What feedback conditioning establishes—and what it does not

Feedback-Conditional Policy (FCP) [5] trains $q_t(x \mid D, f)$ on execution–feedback pairs by conditional maximum likelihood. Its offline Bayesian characterization (Eq. 1) and KL-regularized feedback-likelihood objective (Eq. 2) establish the validity of conditioning as a way to favor executions that are likely to elicit selected feedback. In its *online* phase (Eq. 5 and Algorithm 2), FCP already performs *repeated* desired-feedback-conditioned rollout collection, receives fresh *actual* feedback,

and updates the same conditional model. Thus GFT must not claim to have introduced either the feedback posterior or online feedback-conditioned bootstrapping.

Conditioning on a message f^+ does increase the probability of eliciting that *same message* under exact Bayes conditioning: for $m_p(f^+) > 0$,

$$m_{r_p(\cdot|f^+)}(f^+) = m_p(f^+) + \frac{\mathrm{Var}_{X\sim p}[F(f^+ \mid X)]}{m_p(f^+)} \geq m_p(f^+). \tag{47}$$

This statement is *not* an arbitrary task-success improvement theorem. Section 6.3 provides a calibrated and γ-separating example where one exact population GFT step reduces success from 98.1% to 24.0% while its feedback KL sharply decreases. Conditioning directly on the single desired message f_B in that example drops success further, to approximately 5.67%. This observation does not refute FCP's formal Bayesian/KL identity or its separately stated perfect-verifier special case; it limits an informal inference from "more likely to elicit positive-sounding feedback" to "more likely to solve the task." Our iteration theorem supplies a different, global statement under calibration, coverage, and optimality separation.

10.2 Exact-population online FCP has the same rollout dynamics

Fix D, let $G(f)$ be FCP's *fixed* desired-feedback sampling law $p_{\mathrm{user}}(f \mid D)$, and let $q_t(x \mid f)$ be its online conditional policy. Define the actual distribution of executions sampled for its next feedback round,

$$\rho_t(x) := \sum_f G(f) q_t(x \mid f). \tag{48}$$

The source algorithm trains on actual feedback $f \sim F(\cdot \mid x)$, not on the desired feedback used to generate x. Under exact population conditional MLE and positive required feedback support, this produces the following operator.

Proposition 10.1 (Population equivalence of FCP bootstrapping and GFT)**.** *If FCP's online conditional model is exactly fitted at each round to executions $X \sim \rho_t$ and their actual feedback $F \sim F(\cdot \mid X)$, then on supported feedback*

$$q_{t+1}(x \mid f) = \frac{\rho_t(x) F(f \mid x)}{m_{\rho_t}(f)}, \tag{49}$$

and its desired-feedback-marginal rollout distribution satisfies

$$\rho_{t+1}(x) = \rho_t(x) \sum_f F(f \mid x) \frac{G(f)}{m_{\rho_t}(f)}. \tag{50}$$

For $G = G^\star$, $p_0 = \rho_0$, and the same channel F, this is precisely GFT's population trajectory update (12)*, up to the choice of initial rollout law and indexing. In particular, Theorems 5.2 and 6.3 also apply to* this idealized FCP rollout marginal *when their full assumptions hold.*

Proof. The sampled population joint law is $\rho_t(x)F(f \mid x)$, so conditional MLE recovers its Bayes posterior (49). Marginalizing this conditional over desired feedback G gives (50) by direct substitution. Identifying the respective rollout marginals then yields (12). □

This is an *analysis of an exact-population idealization*, not a result claimed or proved by FCP, and it does not prove convergence of FCP's finite-batch, parameter-shared neural training. In

particular, FCP's user-specified positive-feedback distribution is not generally guaranteed to equal a calibrated $G^\star$. Without calibration, the EM-type feedback-KL monotonicity argument still applies to the exact operator wherever defined, but the comparator-dependent rate and behavioral guarantee above cannot simply be asserted. The proof technique is classical EM; the contribution is specifying statistical feedback conditions that connect repeated likelihood matching to task correctness.

For illustration, if $G = \delta_{f^+}$ is concentrated on one fixed desired message, repeated exact population bootstrapping has the closed form

$$\rho_t(x) = \frac{\rho_0(x)F(f^+ \mid x)^t}{\sum_y \rho_0(y)F(f^+ \mid y)^t}, \tag{51}$$

assuming $m_{\rho_0}(f^+) > 0$. It concentrates on executions of maximum $F(f^+ \mid x)$ *within initial support*, not necessarily on correct executions. If f^+ is a perfect binary success signal and the policy initially covers success, the first conditioning step already restricts support to successful executions; this matches FCP's verifier special case.

10.3 Why distill when ideal feedback is unavailable at deployment?

The population equivalence above is deliberately unfavorable to a claim that the additional distillation step improves the *exact population* operator: it does not. Its motivation is instead *training-time privileged information and request-only deployment*. On training requests $D \sim q_{\text{train}}$, the algorithm can afford to obtain $F^\star \sim G^\star_D$; for a new request D_{test}, that task-specific ideal feedback may be unavailable, expensive to construct, or require executing the task first. GFT transfers the ideal-feedback-conditioned rollout mixture into a clean policy $\pi_\theta(X \mid D)$, which is used for the next training interaction and can also act at test time without any feedback input. Unlike a generic positive slogan, a long request-specific successful-execution narrative or annotated log may not be straightforward to instantiate for an unseen task. FCP may still use a generic positive condition at test time, so a deployment advantage is *conditional on the feedback actually being difficult to supply*; it is not an impossibility claim about FCP or about adding distillation to FCP.

The feedback object can be quite rich: a short critique, a partial requirement report, a multi-paragraph execution assessment, or even an annotated past trajectory. These choices do *not* all satisfy our theorem automatically. The format is unrestricted conceptually, but calibration, required support, posterior realizability, and statistical separation remain substantive assumptions about its *distribution*. In particular, if feedback literally contains the entire generating execution, $f = (x, c)$ with a teacher annotation c, then $r_t(x' \mid f) = \mathbf{1}\{x' = x\}$ for supported pairs. A calibrated ideal-feedback sampler must therefore itself provide annotated *optimal* executions. In that special case, exact Stage-II distillation is simply imitation of the optimal witness mixture in one round, and a direct expert-demonstration/SFT baseline is essential. The argument does not claim that copying is harmful; it exposes the supervision needed to make the ideal sample available. When feedback omits the complete execution or reports task-level requirements, ideal feedback may be sampled without a demonstration, as in Section 7.

Our fixed-request population theorems concern supported training requests and make *no* claim about generalization to unseen D_{test}. We treat transfer across requests by shared neural parameters as an *empirical hypothesis*, not as an additional theorem to be proved here. Likewise, finite-capacity and finite-SGD GFT may differ from FCP: the clean-policy bottleneck can help or hurt approximation and training, and no positive effect is proved. A fair empirical comparison should keep the same training feedback and evaluation requests, contrast FCP (with available test-time desired feedback), one-round request-only distillation, and iterative GFT, and include direct SFT when ideal feedback

reveals demonstrations. Crucially, comparisons should report any test-time feedback-generation cost instead of silently granting either method privileged information unavailable to the other.

11 Relation to retrospective self-distillation (SDPO)

The actual SDPO update. Self-Distillation Policy Optimization (SDPO) [6] samples an attempt y from the current student $\pi_\theta(\cdot \mid D)$, receives *actual* hindsight feedback f, and re-prompts the *same model* with (D, f) to construct a retrospective self-teacher $q_\theta(\cdot \mid D, f)$. The student is trained against this teacher's *next-token distributions along the original sampled prefixes* via reverse-KL distillation, with teacher gradients stopped. In the ideal on-policy, full-logit notation of SDPO's Eq. (1), the per-iteration surrogate is

$$\mathcal{L}_t^{\mathrm{SDPO}}(\vartheta) := \mathbb{E}_{\substack{y\sim\pi_{\theta_t}(\cdot|D)\\ f\sim F_D(\cdot|y)}}\left[\sum_h D_{\mathrm{KL}}\big(\pi_\vartheta(\cdot \mid \mathcal{H}_h, D)\|q_{\theta_t}(\cdot \mid \mathcal{H}_h, D, f)\big)\right], \tag{52}$$

where both the prefix/feedback sampling law and self-teacher are *held fixed during this surrogate update*. SDPO's Proposition 2.1 derives its logit-level policy-gradient form under this stop-gradient convention. Its appendix separately considers an additional prefix-distribution gradient and interprets the fixed-teacher reverse KL as optimizing a teacher-defined implicit reward. SDPO also experiments with regularized/EMA teachers and top-K logit approximations; Eq. (52) is the unregularized full-KL population idealization, *not* the complete implementation used in its experiments.

GFT instead learns a conditional *execution posterior* from (D, X, F) triples, independently draws $F^\star \sim G_D^\star$, samples a *new* execution conditioned on $F^\star$, and distills it into a request-only actor. GFT's exact population objective is the feedback-distribution mismatch $D_{\mathrm{KL}}(G_D^\star \| m_p)$; SDPO does not specify this as the objective of its on-policy student updates. In particular, a statistically informative feedback channel does not by itself guarantee that a retrospectively prompted self-teacher outputs corrections rather than describing which action most likely produced the observed feedback.

Proposition 11.1 (A limitation of population reverse-KL self-distillation)**.** *There is a one-step, two-action feedback environment with γ-optimality-separating stochastic feedback and a single, shared-parameter feedback-conditioned self-teacher for which* exact population minimization *of* (52) *converges to the suboptimal action from every initialization assigning the good action probability strictly between* 0 *and* 1/2*. In the same environment, exact population GFT with calibrated ideal feedback converges to the good action from every positive-coverage initialization.*

Proof. Fix one request and actions g (optimal) and b (suboptimal). Let feedback be $+$ or $-$, with

$$F(+ \mid g) = F(- \mid b) = 0.9, \qquad F(- \mid g) = F(+ \mid b) = 0.1.$$

The symbols denote stochastic feedback reports, not perfectly reliable binary success labels. The statistic $s(f) = \mathbf{1}\{f = +\}$ separates the actions with $a = 0.9$ and $\gamma = 0.8$. Write $u = \pi_\theta(g) = \sigma(\theta)$, $\sigma(z) = (1 + e^{-z})^{-1}$, and $c = \log 9$. Consider a shared-parameter model whose retrospectively conditioned probabilities are

$$q_\theta(g \mid +) = \sigma(\theta + c), \qquad q_\theta(g \mid -) = \sigma(\theta - c).$$

These conditionals are precisely the Bayes posteriors $\pi_\theta(g)F(f \mid g)/m_\theta(f)$, and are representable with the same θ as the clean policy. At round t, freeze these teacher conditionals and the on-policy feedback marginal $m_t(+) = 0.1 + 0.8u_t$, and minimize the one-step instance of (52) exactly. Because

a weighted average of binary reverse KL divergences has minimizer with log odds equal to the weighted average of the teacher log odds,

$$\begin{aligned}\operatorname{logit}(u_{t+1}) &= m_t(+)\operatorname{logit} q_{\theta_t}(g \mid +) + (1 - m_t(+))\operatorname{logit} q_{\theta_t}(g \mid -) \\ &= \operatorname{logit}(u_t) + (1.6u_t - 0.8)\log 9. \end{aligned} \tag{53}$$

If $0 < u_0 < 1/2$, the log odds strictly decrease, the probabilities decrease, and their only possible limit is zero. Hence $u_t \to 0$. On the other hand, calibrated ideal feedback for GFT is $G^\star = F(\cdot \mid g)$; initial positive coverage holds for every $u_0 > 0$, so Theorem 6.3 implies $p_t(g) \to 1$. □

Interpretation and scope. At $u_0 = 0.2$, the exact population reverse-KL self-distillation step in Proposition 11.1 gives $u_1 \approx 0.0801$, whereas GFT's exact operator gives $p_1(g) \approx 0.6258$. The example is not a failure claim about the *empirical* SDPO implementation: the constructed teacher is an exact Bayes *posterior of the attempted action given feedback*, while SDPO intends its re-prompted teacher to use feedback *correctively*. The paper does not claim that its self-teacher is a Bayes posterior. Indeed, SDPO's positive empirical findings demonstrate that retrospective correction can work well, including with optional successful examples in its feedback context. The example establishes only that GFT's calibration and γ-separation assumptions, *without a corrective-teacher condition*, do not imply behavioral convergence for the unregularized reverse-KL SDPO population surrogate.

No overall assumption ordering. GFT requires access to an independently sampled, calibrated $G_D^\star$ and exact population posterior/distillation realizability; SDPO does not require such a generator but depends on the retrospective model's usefulness as a corrective teacher. Its published analysis derives a distillation gradient, trust-region teacher construction, and an implicit-reward interpretation; it does not establish a general global behavioral-convergence theorem for its actual online iterates. The assumptions are not ordered by strength. For example, in the two-action special case, a sufficient *additional* condition for exact population SDPO improvement is that the average self-teacher log odds of g exceed the current student log odds; this concerns the teacher's corrective quality, not merely the statistical information in F. Neither this condition nor Proposition 11.1 should be attributed as a theorem from [6].

12 Extensions, failure examples, and scope

Distributional separation, not merely pairwise separation. Let one optimal execution produce "A" and "B" with equal probability, while two suboptimal executions produce "A" and "B" deterministically, respectively. Each bad execution has feedback different from the optimal execution, but the equal mixture of the two bad executions matches ideal feedback exactly. The score condition excludes this mixture-level aliasing.

Task averaging. The above results are conditional on a fixed request. If $D \sim q^\star$, shared-policy exact population learning induces the request-wise update for $q^\star$-almost every D under the stated conditional realizability assumptions (and at every positive-mass D when $q^\star$ is discrete). For request-dependent $\gamma_D > 0$, average binary error is bounded by $\mathbb{E}_{D\sim q^\star}[\gamma_D^{-1}\sqrt{V_{0,D}/(2(T+1))}]$, provided the expectation is finite. If $\gamma_D \geq \gamma_0 > 0$, Jensen yields $\mathbb{E}_D p_T(\mathcal{C}(D)^c \mid D) \leq \gamma_0^{-1}\sqrt{\mathbb{E}_D V_{0,D}/(2(T+1))}$. No gradient or representation sample-complexity bound is implied by this population averaging.

Approximate ideal-feedback calibration: a qualified extension. Suppose the available generator $\widetilde{G}$ is within ϵ_G total variation of $G^\star$ and is itself realizable as $\widetilde{G} = F\widetilde{p}$ for a distribution $\widetilde{p}$ covered by p_0 (not necessarily optimal). Exact population GFT using $\widetilde{G}$ obeys Theorem 5.2 with

comparator $\widetilde{p}$. Under the original γ-separation of $G^\star$, the triangle inequality gives

$$p_T(\mathcal{C}^c) \le \frac{1}{\gamma}\left[\epsilon_G + \sqrt{\frac{D_{\mathrm{KL}}(\widetilde{p}\|p_0)}{2(T+1)}}\right]. \tag{54}$$

This is not a guarantee for an arbitrary unachievable generator $\widetilde{G}$: its realizability and coverage are additional assumptions.

Why access to $G^\star$ matters. The ability to sample plausible positive task-relative feedback before observing a successful trajectory is valuable supervision. An LLM or task verifier generating "all requested conditions were met" may be easy to prompt but difficult to calibrate faithfully. The population theorem cannot establish a fair sample-efficiency advantage over a scalar learner unless this access and its construction cost are accounted for.

Support and first-success limitations. If p_0 assigns zero probability to every optimal execution, the multiplicative update cannot create optimal support. In the deterministic full-audit limit, conditioning on wholly positive feedback is equivalent to selecting successful trajectories; a finite dataset may never contain one. Random partial audits permit overlap of feedback between correct and incorrect executions, but the present theorem does not establish an efficient finite-sample learning rate for that case. The exact-population proof is therefore not an end-to-end polynomial-interaction guarantee.

Representations, modalities, and intentions. Stage-I posterior realizability and Stage-II distillation realizability are strong population conditions. Natural-language teachers may omit crucial requirements, paraphrases may be difficult to ground, and other modalities may require expensive encoders or discretization. GFT does not explicitly interpret feedback as a request to change behavior: a critique may identify a mistake without the posterior model ever applying its suggested correction. The bijective relabeling property concerns the exact operator, not sample efficiency or neural generalization. A fixed neural architecture may be unable to represent the posterior. The finite-tabular batch theorem in Section 8.1 controls empirical conditional-MLE and sampling errors but has potentially enormous constants; approximate neural optimization still requires additional analysis. Our finite-execution theorem does not establish convergence of neural parameters, generalization to unobserved requests, or identification of a particular optimal mixture.

13 Conclusion

We study General Feedback Training (GFT) through an explicit objective: match the desired feedback distribution to the law induced by executing a request-only policy. The formal update is modality- and communicative-intention-agnostic: it treats feedback as an observation of behavior rather than an instruction to enact. The resulting population EM operator has a decreasing feedback-KL potential and a quantitative last-iterate rate. Statistical feedback separation identifies when this implies correct behavior, while a calibrated counterexample shows why task success need not improve each round. Truthful partial requirement reports satisfy the separation condition, and a common optimal-feedback law additionally makes success monotone. Our comparison to the language-feedback algorithm FCP shows that the exact population rollout operator is not unique to distillation; the distinct practical role of GFT's student is to act without privileged, task-specific feedback at deployment.

For one sampled batch algorithm, we prove two deliberately different results: a high-probability behavioral rate for unrestricted finite-tabular empirical fits, and constrained stationary-point convergence for bounded, posterior-realizable linear-softmax policies under strong curvature, growing

batches, and controlled empirical optimization error. Neither result guarantees efficient horizon-dependent sample complexity, global optimality for linear-softmax policies, arbitrary neural optimization, or generalization to unseen requests. These boundaries guide the empirical comparison with FCP and direct supervised learning from any available expert demonstrations.

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A Details of the finite-tabular batch convergence theorem

We prove Theorem 8.1 for one request and make explicit how sampled conditional imitation and student distillation produce the needed approximation to the population operator. Here $P_t(x, f) = p_t(x)F(f \mid x)$ denotes the *current actual* joint law, m_t its feedback marginal, and $q_t = \mathcal{T}(p_t)$ the exact population update.

1. Two concentration inequalities and the learned posterior. For any distribution P on a set of d elements and its empirical distribution $\widehat{P}$ based on N independent samples, applying Hoeffding's inequality to each subset and taking a union bound gives

$$\Pr\Big(\mathrm{TV}(\widehat{P}, P) > u\Big) \le 2^{d+1} e^{-2Nu^2}. \tag{55}$$

At round t define $\widehat{m}_t(f) = \sum_x \widehat{P}_t(x, f)$ and $\widehat{r}_t(x \mid f) = \widehat{P}_t(x, f)/\widehat{m}_t(f)$ when the denominator is positive. On unseen feedback, choose any conditional probability vector. Because $\widehat{P}_t(x, f) = \widehat{m}_t(f)\widehat{r}_t(x \mid f)$ even when $\widehat{m}_t(f) = 0$,

$$\begin{aligned}\sum_f m_t(f)\,\mathrm{TV}(r_t(\cdot \mid f), \widehat{r}_t(\cdot \mid f)) &\le \mathrm{TV}(P_t, \widehat{P}_t) + \mathrm{TV}(m_t, \widehat{m}_t) \\ &\le 2\,\mathrm{TV}(P_t, \widehat{P}_t).\end{aligned}$$

If $p_t(x) \ge b$ for each $x \in S = \mathrm{supp}(\bar{p})$, calibration ensures

$$m_t(f) = \sum_x p_t(x) F(f \mid x) \ge b \sum_{x \in S} \bar{p}(x) F(f \mid x) = b G^\star(f). \tag{56}$$

Hence, with $\widehat{q}_t = \sum_f G^\star(f)\widehat{r}_t(\cdot \mid f)$,

$$\begin{aligned}\mathrm{TV}(q_t, \widehat{q}_t) &\le \sum_f G^\star(f)\,\mathrm{TV}(r_t(\cdot \mid f), \widehat{r}_t(\cdot \mid f)) \\ &\le \frac{2}{b}\,\mathrm{TV}(P_t, \widehat{P}_t).\end{aligned} \tag{57}$$

The second batch consists of independent trajectories with law $\widehat{q}_t$ conditional on the fitted auxiliary model; exact empirical trajectory MLE therefore yields $p_{t+1} = \widehat{p}_t^{+}$, the empirical law of these M_t pseudo-executions. Applying (55) to the nk-point actual joint and the n-point pseudo-execution law, with per-stage failure probability $\delta/[2(t+1)(t+2)]$, and using (32), gives

$$\mathrm{TV}(q_t, \widehat{q}_t) \le \frac{e_t}{2}, \qquad \mathrm{TV}(\widehat{q}_t, p_{t+1}) \le \frac{e_t}{2}. \tag{58}$$

The bound is conditional on the past and on $p_t(x) \ge b$ for $x \in S$. The sum of the per-round failure probabilities is at most δ because $\sum_{t \ge 0} 1/[(t+1)(t+2)] = 1$. The event of successful sampling in every round therefore has probability at least $1 - \delta$, as long as we prove the required lower bound inductively.

2. Support is not lost on optimal witness trajectories. Write $V(p) = D_{\mathrm{KL}}(\bar{p} \| p)$ and $h = -\sum_{x \in S} \bar{p}(x) \log \bar{p}(x)$. If $V(p_t) \le v + 1$, then

$$\sum_{x \in S} \bar{p}(x)[-\log p_t(x)] = V(p_t) + h \le v + 1 + h.$$

Every term on the left is nonnegative, and $\bar{p}(x) \ge a$, so $p_t(x) \ge b = \exp[-(v + 1 + h)/a]$ for every $x \in S$. For such an x, $G^\star(f) \ge \bar{p}(x)F(f \mid x)$, $m_t(f) \le 1$, and Cauchy–Schwarz gives $\sum_f F(f \mid x)^2 \ge 1/k$. Thus

$$\begin{aligned}q_t(x) &= p_t(x) \sum_f F(f \mid x) \frac{G^\star(f)}{m_t(f)} \\ &\ge p_t(x)\bar{p}(x) \sum_f F(f \mid x)^2 \ge \frac{ba}{k} = 2\beta.\end{aligned} \tag{59}$$

On the event (58), $\mathrm{TV}(p_{t+1}, q_t) \le e_t \le \beta/4$, so for each $x \in S$, $|p_{t+1}(x) - q_t(x)| \le e_t$ and $p_{t+1}(x) \ge \beta$. Crucially, no lower bound is imposed on incorrect-execution probabilities; they may converge to zero.

3. Stability of the two KL potentials. For any two distributions p, q satisfying $p(x), q(x) \ge \beta$ on S, the mean-value theorem for log gives

$$|V(p) - V(q)| \le \frac{\|p - q\|_1}{\beta} = \frac{2}{\beta}\,\mathrm{TV}(p, q). \tag{60}$$

Moreover, for every f with $G^\star(f) > 0$, calibration gives $m_p(f), m_q(f) \ge \beta G^\star(f)$, so

$$|D_{\mathrm{KL}}(G^\star \| m_p) - D_{\mathrm{KL}}(G^\star \| m_q)| \le \frac{\|m_p - m_q\|_1}{\beta} \le \frac{2}{\beta}\,\mathrm{TV}(p, q). \tag{61}$$

The exact population inequalities (14) applied between p_t and q_t therefore imply, on the sampling event,

$$V(p_t) - V(p_{t+1}) \ge \Phi_t - \frac{2e_t}{\beta}, \qquad \Phi_{t+1} \le \Phi_t + \frac{2e_t}{\beta}. \tag{62}$$

Since $\sum_{t \ge 0} (t+1)^{-3} \le 3/2$,

$$\frac{2}{\beta} \sum_{t \ge 0} e_t \le \frac{1}{2}\left(\frac{3}{2}\right) = \frac{3}{4} < 1.$$

Starting at $V(p_0) = v$, induction using (62) shows $V(p_t) \le v + 3/4 < v + 1$ for every round. This closes the inductive support and conditional-concentration arguments.

4. Last-iterate feedback KL and behavioral optimality. Summing the first inequality in (62) yields $\sum_{t=0}^{T} \Phi_t \le v + (2/\beta) \sum_{t=0}^{T} e_t$. The second inequality gives $\Phi_T \le \Phi_t + (2/\beta) \sum_{j=t}^{T-1} e_j$ for $0 \le t \le T$. Averaging over t and combining the inequalities,

$$(T+1)\Phi_T \le v + \frac{2}{\beta} \sum_{t=0}^{T} (t+2) e_t. \tag{63}$$

Now $e_t = \beta/[4(t+1)^3]$ and

$$\sum_{t=0}^{\infty} \frac{t+2}{(t+1)^3} = \sum_{j=1}^{\infty} \left(\frac{1}{j^2} + \frac{1}{j^3}\right) \le 2 + \frac{3}{2} = \frac{7}{2}.$$

Thus (63) implies (34). The same good sampling event holds over all times, so the bound holds simultaneously for all T. Under γ-optimality separation, Lemma 6.2 gives $\gamma p_T(\mathcal{C}^c) \le \mathrm{TV}(G^\star, m_{p_T})$; Pinsker and (34) give (35). □

Limitations. The theorem uses unrestricted tables indexed by complete executions and feedback strings and exact empirical fits, not restricted neural models or approximate SGD. The high-probability schedule depends on inaccessible comparator quantities $\bar{p}$ and v (or conservative known bounds on them). The inverse-coverage constant b can be extraordinarily small; $n = |\mathcal{X}(D)|$ and $k = |\mathcal{F}(D)|$ can be exponential in problem size. The result establishes existence of an explicit finite-sample schedule, not efficient implementability or generalization across requests.

B Proof of finite-batch linear-softmax stationarity

We prove Theorem 8.2 for the two sampled fits in Algorithm 2. The argument allows changing θ_t and an auxiliary model fitted afresh on each round; it does *not* assume that sampled student trajectories come from an exact posterior.

Step 1: posterior error from conditional empirical MLE. Let $P_\theta(D, x, f) = q^\star(D)p_\theta(x \mid D)F_D(f \mid x)$ and let $\ell_\theta(\phi) = \mathbb{E}_{P_\theta}[-\log p_\phi^{\mathrm{fb}}(X \mid D, f)]$. In the deterministic finite-horizon environment the trajectory log likelihood is the sum of action log probabilities, which is exactly the behavior-cloning loss in Stage I. Posterior realizability gives

$$\ell_\theta(\phi) - \ell_\theta(\phi^\star(\theta)) = \mathbb{E}_{D\sim q^\star,\, f\sim m_\theta(\cdot|D)} D_{\mathrm{KL}}\Big(r_\theta(\cdot \mid D, f)\Big\|p_\phi^{\mathrm{fb}}(\cdot \mid D, f)\Big). \tag{64}$$

On the empirical deviation event, an empirical NLL minimizer up to a_t satisfies

$$\ell_{\theta_t}(\phi_t) - \ell_{\theta_t}(\phi^\star(\theta_t)) \le 2u_t + a_t. \tag{65}$$

Write $d_t(D, f) = \mathrm{TV}(r_{\theta_t}(\cdot \mid D, f), p_{\phi_t}^{\mathrm{fb}}(\cdot \mid D, f))$. Since $G_D^\star(f) \le \kappa m_{\theta_t}(f \mid D)$ and $\mathrm{TV}(P, Q) \le \sqrt{D_{\mathrm{KL}}(P\|Q)/2}$, Jensen and (64)–(65) give

$$\mathbb{E}_{D\sim q^\star,\, f\sim G_D^\star} d_t(D, f) \le \sqrt{\frac{1}{2}\mathbb{E}_{D,f\sim G_D^\star} D_{\mathrm{KL}}(r_{\theta_t}\|p_{\phi_t}^{\mathrm{fb}})} \le \sqrt{\frac{\kappa(2u_t + a_t)}{2}} =: \rho_t. \tag{66}$$

Step 2: approximate EM maximization after sampled distillation. Recall $Q_{\theta_t}(v)$ from (37) and define $\widetilde{Q}_t(v) = \mathbb{E}_{D,f\sim G_D^\star,\, x\sim p_{\phi_t}^{\mathrm{fb}}(\cdot|D,f)} \log p_v(x \mid D)$. If $|\log p_v(x \mid D)| \le L_x$, the total-variation dual bound gives uniformly in $v \in \Theta$

$$|Q_{\theta_t}(v) - \widetilde{Q}_t(v)| \le 2L_x\rho_t. \tag{67}$$

The second sampling deviation is at most z_t, so $\sup_v |Q_{\theta_t}(v) - \widehat{Q}_t(v)| \le \Delta_t := 2L_x\rho_t + z_t$. Let $v_t^\star \in \arg\max_{v\in\Theta} Q_{\theta_t}(v)$. Since the empirical student is within b_t of the maximum of $\widehat{Q}_t$,

$$\begin{aligned} Q_{\theta_t}(\theta_{t+1}) &\ge \widehat{Q}_t(\theta_{t+1}) - \Delta_t \ge \widehat{Q}_t(v_t^\star) - b_t - \Delta_t \\ &\ge Q_{\theta_t}(v_t^\star) - 2\Delta_t - b_t = Q_{\theta_t}(v_t^\star) - \eta_t. \end{aligned} \tag{68}$$

In particular, the clean M-step is an η_t-approximate maximizer of the *true* EM surrogate.

Step 3: EM ascent and square-summable parameter increments. For any $u, v \in \Theta$, the Gibbs variational identity applied to each desired feedback message yields

$$J(v) - J(u) \ge Q_u(v) - Q_u(u). \tag{69}$$

Indeed, the missing terms in the two corresponding likelihood lower bounds are the nonnegative posterior KL at the new parameter, while the lower bound is tight at u. By μ-strong concavity of Q_{θ_t}, for the midpoint $w = (\theta_t + \theta_{t+1})/2 \in \Theta$,

$$Q_{\theta_t}(w) \ge \tfrac{1}{2}Q_{\theta_t}(\theta_t) + \tfrac{1}{2}Q_{\theta_t}(\theta_{t+1}) + \tfrac{\mu}{8}\|\theta_{t+1} - \theta_t\|^2.$$

Approximate maximization in (68) gives $Q_{\theta_t}(\theta_{t+1}) \ge Q_{\theta_t}(w) - \eta_t$. Rearranging and using (69),

$$J(\theta_{t+1}) - J(\theta_t) \ge \tfrac{\mu}{4}\|\theta_{t+1} - \theta_t\|^2 - 2\eta_t. \tag{70}$$

Because $J(\theta) = -\mathbb{E}_D D_{\mathrm{KL}}(G_D^\star\|m_\theta) \le 0$ and $\sum_t \eta_t < \infty$, summation proves $\sum_t \|\theta_{t+1} - \theta_t\|^2 < \infty$ and hence $\|\theta_{t+1} - \theta_t\| \to 0$.

Step 4: constrained stationary points. Strong concavity and the first-order optimality of $v_t^\star$ imply

$$\tfrac{\mu}{2}\|\theta_{t+1} - v_t^\star\|^2 \le Q_{\theta_t}(v_t^\star) - Q_{\theta_t}(\theta_{t+1}) \le \eta_t. \tag{71}$$

Thus $\|\theta_t - v_t^\star\| \le \|\theta_t - \theta_{t+1}\| + \sqrt{2\eta_t/\mu} \to 0$. The exact constrained maximizer obeys $v_t^\star = \Pi_\Theta(v_t^\star + \nabla_v Q_{\theta_t}(v_t^\star))$. On the other hand, differentiating through the feedback marginal (the Fisher identity from Proposition 9.1) yields

$$\nabla J(\theta_t) = \nabla_v Q_{\theta_t}(v)\big|_{v=\theta_t}. \tag{72}$$

Nonexpansiveness of Π_Θ and the uniform L_Q gradient Lipschitz bound now give

$$\begin{aligned}
&\|\theta_t - \Pi_\Theta(\theta_t + \nabla J(\theta_t))\| \\
&\quad \le 2\|\theta_t - v_t^\star\| + \|\nabla_v Q_{\theta_t}(\theta_t) - \nabla_v Q_{\theta_t}(v_t^\star)\| \\
&\quad \le (2 + L_Q)\|\theta_t - v_t^\star\| \longrightarrow 0.
\end{aligned}$$

Compactness of Θ and continuity of the projected-gradient mapping establish constrained stationarity of every accumulation point. If every sufficiently late iterate is separated from the boundary by a fixed margin, the bounded gradient and the vanishing projected-gradient mapping give the ordinary gradient limit.

Step 5: making the empirical deviations hold with high probability. We spell out a sufficient, deliberately conservative, sample-size construction. For either supervised loss family $h_w(Z) \in [0, L]$ with parameter $w \in W \subseteq \{\|w\| \le R\} \subset \mathbb{R}^d$, assume $h_w(Z)$ is Λ-Lipschitz in w for every data item. An $\varepsilon/(4\Lambda)$ covering net for W has at most $(1 + 8R\Lambda/\varepsilon)^d$ members. Hoeffding's inequality at the net points and the Lipschitz extension show

$$\Pr\left(\sup_{w\in W} |\widehat{L}_N(w) - L(w)| > \varepsilon\right) \le 2\left(1 + \frac{8R\Lambda}{\varepsilon}\right)^d \exp\left(-\frac{N\varepsilon^2}{2L^2}\right). \tag{73}$$

This conditional concentration inequality applies to Stage I given the current θ_t, and to Stage II given both θ_t and the fitted ϕ_t. Put $u_t = c(t+1)^{-4}$, $z_t = c(t+1)^{-2}$, $a_t \le c(t+1)^{-4}$, and $b_t \le c(t+1)^{-2}$ for any fixed $c > 0$. Choose N_t and M_t by solving (73) with the respective dimensions and loss bounds, and individual failure probabilities $\delta/[2(t+1)(t+2)]$. This gives

$$N_t = O\big((t+1)^8[d_{\mathrm{fb}}\log(t+2) + \log((t+2)/\delta)]\big), \quad M_t = O\big((t+1)^4[d_\theta \log(t+2) + \log((t+2)/\delta)]\big),$$

up to fixed compactness and loss constants. A conditional union bound makes both deviation events hold for every t with probability at least $1 - \delta$. On this event, and assuming the separately stipulated optimization-error tolerances, the resulting η_t from (40) is $O((t+1)^{-2})$, so its series is summable. This proves the claimed high-probability stationarity for the constrained, approximately globally fitted linear-softmax batch specialization. The concentration calculation does not bound the computation needed to achieve the optimization tolerances. □

C Supplementary geometric view of the population update

The exact update has a Fisher–Rao interpretation on the full trajectory simplex, but this does not imply that finite-parameter neural SGD is a natural gradient step. We retain the result for completeness.

C.1 Exact natural gradient on the trajectory simplex

There is also a geometric interpretation of the original GFT update. For a fixed request and an execution distribution p in the relative interior of its supported face, define

$$W_p(x) := \sum_f F(f \mid x)\frac{G^\star(f)}{m_p(f)}, \qquad J(p) := -D_{\mathrm{KL}}(G^\star \| m_p).$$

Then $\partial J/\partial p(x) = W_p(x)$ and $\sum_x p(x)W_p(x) = 1$. The Fisher–Rao natural-gradient tangent vector on the trajectory simplex is

$$(\widetilde{\nabla} J(p))(x) = p(x)\big(W_p(x) - 1\big). \tag{74}$$

Thus the exact GFT operator from Lemma 5.1 is precisely the *unit explicit-Euler step* $\mathcal{T}(p) = p + \widetilde{\nabla} J(p)$. This is not an identity between neural-parameter SGD and Fisher–Rao natural gradient.

Proposition C.1 (Convergence of exact damped trajectory-space GFT)**.** *Under the original exact-population assumptions of Theorem 5.2, let $\bar{p}$ be a calibrated, initially covered optimal witness. Fix $0 < \eta \le 1$ and define*

$$p_{t+1} = (1-\eta)p_t + \eta\mathcal{T}(p_t), \quad \Phi_t = D_{\mathrm{KL}}(G^\star \| m_{p_t}), \quad V_t = D_{\mathrm{KL}}(\bar{p} \| p_t).$$

Then

$$\Phi_{t+1} \le \Phi_t, \quad V_t - V_{t+1} \ge \eta\Phi_t, \quad \Phi_T \le \frac{V_0}{\eta(T+1)}. \tag{75}$$

If the original γ-optimality-separating condition holds, then

$$p_T(\mathcal{C}^c) \le \frac{1}{\gamma}\sqrt{\frac{V_0}{2\eta(T+1)}}. \tag{76}$$

Proof. Both $p \mapsto D_{\mathrm{KL}}(G^\star \| m_p)$ and $p \mapsto D_{\mathrm{KL}}(\bar{p} \| p)$ are convex. Hence each potential at p_{t+1} is at most $(1-\eta)$ times its value at p_t plus η times its value at $\mathcal{T}(p_t)$. Theorem 5.2 gives $\Phi(\mathcal{T}(p_t)) \le \Phi_t$ and $V_t - V(\mathcal{T}(p_t)) \ge \Phi_t$. Summing the second inequality and using the first to make Φ_t nonincreasing gives (75). The original feedback-statistic bound and Pinsker yield (76). □

Scope. The damped update operates on the *full trajectory simplex*; it may require exponentially large representations. It does not show that an ordinary neural step realizes this update, nor that its gradients can be estimated efficiently.

D Online GFT: incremental SGD for both policies

We next replace full inner-loop optimization of the two supervised stages by continuous stochastic gradients, while preserving the original policy roles and access to feedback. The resulting convergence theorem is *conditional*: the auxiliary loss is required to have optimization geometry strong enough to track its moving posterior, and the clean policy is shown to reach stationary points, not necessarily globally correct behavior.

D.1 Algorithm and supervised auxiliary objective

For differentiable auxiliary parameters ϕ, let $r_\phi(x \mid D, f)$ be the *executed trajectory law* of $\pi_\phi(a \mid \mathcal{H}, D, f)$ in the unchanged deterministic environment. Its population conditional behavioral-cloning loss is

$$\ell(\phi;\theta) := \mathbb{E}_{\substack{D\sim q^\star,\, X\sim p_\theta(\cdot|D) \\ F\sim F_D(\cdot|X)}}[-\log r_\phi(X \mid D, F)]. \tag{77}$$

The loss is just negative Stage-I conditional likelihood from Equation (9), expressed at the trajectory level. No feedback parser, scalar reward, or action-alignment oracle is introduced.

Algorithm 3 Online GFT: ordinary SGD for both policies

Input: $q^\star$, F_D, $G^\star_D$; clean parameters θ_0; auxiliary parameters ϕ_0; step sizes $\{\alpha_t, \beta_t\}$

1: **for** $t = 0, 1, 2, \ldots$ **do**
2: Sample $D_t \sim q^\star$, roll out π_{θ_t} to obtain X_t, and obtain hindsight feedback $F_t \sim F_{D_t}(\cdot \mid X_t)$.
3: $\widehat{\ell}^{\mathrm{fb}}_t(\phi) \leftarrow -\sum_{h=1}^{H} \log \pi^{\mathrm{fb}}_\phi(a_h^{(t)} \mid \mathcal{H}_h^{(t)}, D_t, F_t)$.
4: **SGD:** $\phi_{t+1} \leftarrow \phi_t - \beta_t \nabla_\phi \widehat{\ell}^{\mathrm{fb}}_t(\phi_t)$.
5: Independently sample $D'_t \sim q^\star$ and ideal feedback $F^\star_t \sim G^\star_{D'_t}$.
6: Roll out $\pi^{\mathrm{fb}}_{\phi_{t+1}}(\cdot \mid \mathcal{H}_h, D'_t, F^\star_t)$ to obtain X^+_t.
7: $\widehat{\ell}^{\mathrm{clean}}_t(\theta) \leftarrow -\sum_{h=1}^{H} \log \pi_\theta(a_h^{+,(t)} \mid \mathcal{H}_h^{+,(t)}, D'_t)$.
8: **SGD:** $\theta_{t+1} \leftarrow \theta_t - \alpha_t \nabla_\theta \widehat{\ell}^{\mathrm{clean}}_t(\theta_t)$ ▷ hold X^+_t fixed
9: **end for**

Output: Continuously deployable clean policy π_{θ_t}

Algorithm 3 uses *ordinary stochastic gradient descent* on two negative log-likelihood losses; each iteration uses one freshly sampled execution–feedback triple and one freshly sampled ideal-feedback-conditioned execution (or independent minibatches). The first update trains the auxiliary model to imitate actions conditional on hindsight feedback; the second trains the request-only policy to imitate the generated execution. The generated execution X^+_t is held fixed when differentiating the clean-policy loss. The trajectory law $p_{\theta_t}(\cdot \mid D)$ denotes rollouts of π_{θ_t}, not a separately maintained density model. Auxiliary data must reflect the *current* clean-policy joint: naive mixing of old, unweighted replay data changes the population loss.

Projection in the theorem, not the practical pseudocode. To control the auxiliary iterates, Theorem D.1 studies the *projected* variant of the auxiliary SGD step, $\phi_{t+1} = \Pi_K[\phi_t - \beta_t \nabla_\phi \widehat{\ell}^{\mathrm{fb}}_t(\phi_t)]$, for a compact convex K. Theorem D.1 applies to Algorithm 3 as written only when the unprojected auxiliary iterates remain in K (so projection has no effect) and all the theorem's other assumptions hold. It does not automatically guarantee convergence for unconstrained neural-network SGD.

D.2 A conditional two-timescale convergence theorem

The following assumptions go beyond the original exact-population GFT theorem. They are sufficient for the projected auxiliary-SGD variant of Algorithm 3; for unprojected SGD they suffice only if the projection would remain inactive along its trajectory. These are not claims about all neural language models or about Section 7's partial-audit construction.

(A1) Smooth bounded actor and compatible support. The unprojected actor iterates stay almost surely in a compact region Θ_0, and J has an L_J-Lipschitz gradient on a convex neighborhood containing every update segment. For each parameter pair visited, all trajectories

that can be sampled from the ideal-feedback-conditioned auxiliary policy have positive clean-policy probability, so their clean-policy scores are well defined and obey $\|s_\theta(x, D)\| \le B$. The same score bound holds on the current clean-policy support, and ideal-supported feedback has positive $m_\theta(f \mid D)$. A full-support, uniformly score-bounded softmax actor is one sufficient way to meet the score-support requirements.

(A2) Strongly convex posterior tracking. Auxiliary parameters lie in a compact convex set K. For every $\theta \in \Theta_0$, $\ell(\cdot;\theta)$ is differentiable and uniformly μ-strongly convex on K, with a unique minimizer $\phi^\star(\theta) \in K$ exactly realizing $r_\theta(\cdot \mid D, f)$ for every ideal-supported (D, f). The minimizer map is $L_\star$-Lipschitz.

(A3) Sampling regularity. The auxiliary stochastic gradient $\widehat{h}_t$ is conditionally unbiased for $\nabla_\phi \ell(\phi_t; \theta_t)$, with conditional second moment at most M^2. Furthermore, $\delta(\theta, \phi) \le L_r \|\phi - \phi^\star(\theta)\|$, and the actor uses conditionally fresh ideal-feedback rollouts, with score norm at most B.

(A4) Separated steps. Positive deterministic steps α_t, β_t satisfy

$$\sum_t \alpha_t = \sum_t \beta_t = \infty, \quad \sum_t (\alpha_t^2 + \beta_t^2) < \infty, \quad \frac{\alpha_t}{\beta_t} \to 0, \quad \sum_t \frac{\alpha_t^2}{\beta_t} < \infty. \tag{78}$$

For example $\beta_t \asymp (t+1)^{-0.6}$ and $\alpha_t \asymp (t+1)^{-0.9}$ satisfy these conditions.

Theorem D.1 (Almost-sure stationarity of projected two-timescale Online GFT)**.** *Under assumptions (A1)–(A4) above, define $\widehat{h}_t := \nabla_\phi \widehat{\ell}_t^{\mathrm{fb}}(\phi_t)$, where $\widehat{\ell}_t^{\mathrm{fb}}$ is the per-episode loss in Algorithm 3. The* projected *auxiliary-SGD update and the same clean-policy SGD update,*

$$\phi_{t+1} = \Pi_K \left[\phi_t - \beta_t \widehat{h}_t\right], \qquad \theta_{t+1} = \theta_t + \alpha_t s_{\theta_t}(X_t^+, D_t'),$$

satisfy, almost surely,

$$\|\phi_t - \phi^\star(\theta_t)\| \longrightarrow 0, \tag{79}$$

$$\sum_t \beta_t \|\phi_t - \phi^\star(\theta_t)\|^2 < \infty, \tag{80}$$

$$\sum_t \alpha_t \|\nabla J(\theta_t)\|^2 < \infty, \tag{81}$$

$$\|\nabla J(\theta_t)\| \longrightarrow 0. \tag{82}$$

Thus every limit point of the clean-policy parameters is stationary for J. The theorem does not *establish $p_{\theta_t}(\mathcal{C}(D) \mid D) \to 1$, monotonic task success, or polynomial interaction complexity.*

Proof. The proof is given in Appendix E. Its two steps are posterior tracking by a Robbins–Siegmund drift argument, followed by stochastic ascent with a summable gradient-bias term. The conditions are stated in full because the original population assumptions alone do not imply them. □

D.3 What the result does and does not say

In a finite tabular conditional model with positive trajectory and feedback probabilities, direct probability-vector auxiliary parameters in a compact product of simplexes give a uniformly strongly convex population negative conditional log-likelihood: its Hessian is diagonal with entries proportional to positive joint probabilities divided by squared model probabilities. This verifies that the auxiliary assumptions are *nonvacuous*, but the representation may be exponentially large.

For truthful partial audits from Section 7, many feedback messages have zero likelihood after incompatible executions. The same interior and uniform-strong-convexity argument is not automatic; neither raw-language neural conditional imitation nor efficient posterior estimation is proved by the theorem.

Finally, a restricted actor class may have stationary policies that are not optimal even when feedback is perfectly separating. For a one-step success/failure task with deterministic binary feedback and ideal feedback always equal to "success," set $p_\theta(\text{success}) = \sigma(\theta^2)$. Then $J(\theta) = \log \sigma(\theta^2)$, so $\nabla J(0) = 0$ while success probability is only $1/2$. Hence the original GFT optimality-separation condition alone cannot upgrade neural-parameter stationarity to task success.

Research boundary. The original exact GFT operator achieves global feedback matching and, under separation, correct behavior. The projected auxiliary-SGD version of Online GFT achieves a more conventional *stationary-point* guarantee under additional smoothness, trackability, and timescale conditions; the unprojected version is covered only when the projection would stay inactive. Establishing posterior tracking for realistic free-form language using polynomially many sampled interactions, and eliminating suboptimal neural stationary points, remain separate open questions.

E Detailed proof of Online GFT convergence

We give the complete stochastic-approximation argument, keeping the two stochastic learning stages and their conditioning separate. Let $\mathcal{F}_t$ be the sigma-field generated by all randomness before the stage-I sample at round t. In particular, θ_t and ϕ_t are $\mathcal{F}_t$-measurable. Define

$$e_t := \phi_t - \phi^\star(\theta_t), \qquad \widetilde{e}_{t+1} := \phi_{t+1} - \phi^\star(\theta_t).$$

Step 1: tracking the moving auxiliary optimum. Projection onto K is nonexpansive; its population minimizer obeys the first-order variational inequality and, by μ-strong convexity,

$$\langle \phi_t - \phi^\star(\theta_t), \nabla_\phi \ell(\phi_t; \theta_t) \rangle \geq \mu \|e_t\|^2.$$

Consequently conditional unbiasedness and the second-moment bound give

$$\mathbb{E}[\|\widetilde{e}_{t+1}\|^2 \mid \mathcal{F}_t] \leq (1 - 2\mu\beta_t)\|e_t\|^2 + M^2\beta_t^2. \tag{83}$$

The actor moves by at most $B\alpha_t$ pathwise and the target minimizer is $L_\star$-Lipschitz, so $\|\phi^\star(\theta_{t+1}) - \phi^\star(\theta_t)\| \leq L_\star B\alpha_t$. Applying $\|u+v\|^2 \leq (1+c\beta_t)\|u\|^2 + (1+1/(c\beta_t))\|v\|^2$ with $c = \mu/2$, for all sufficiently large t we obtain

$$\mathbb{E}[\|e_{t+1}\|^2 \mid \mathcal{F}_t] \leq (1 - \mu\beta_t)\|e_t\|^2 + C_1\beta_t^2 + C_2\frac{\alpha_t^2}{\beta_t}. \tag{84}$$

This inequality does not require independent stage-I and stage-II *targets*; the pathwise actor-step bound suffices. The Robbins–Siegmund almost-supermartingale lemma [9] and (A4) imply that $\|e_t\|^2$ converges and $\sum_t \beta_t \|e_t\|^2 < \infty$ almost surely. Since $\sum_t \beta_t = \infty$, its limit must be zero. This proves (79) and (80).

Step 2: bounding the actor-gradient bias. After stage I, define the actor-score conditional mean and deviation

$$g(\theta_t, \phi_{t+1}) = \mathbb{E}[s_{\theta_t}(X_t^+, D_t') \mid \mathcal{F}_t, \phi_{t+1}], \quad b_t = g(\theta_t, \phi_{t+1}) - \nabla J(\theta_t).$$

Proposition 9.2, (A3), and (83) give, for a constant C,

$$\mathbb{E}[\|b_t\|^2 \mid \mathcal{F}_t] \le 4B^2 L_r^2 \mathbb{E}[\|\widetilde{e}_{t+1}\|^2 \mid \mathcal{F}_t] \le C(\|e_t\|^2 + \beta_t^2). \tag{85}$$

As $\alpha_t \le \beta_t$ eventually, the tracking sum and square-summable steps imply

$$\sum_t \alpha_t \mathbb{E}[\|b_t\|^2 \mid \mathcal{F}_t] < \infty \qquad \text{almost surely.} \tag{86}$$

Step 3: stochastic ascent and stationary gradients. Use the smooth ascent inequality on each actor update segment, condition first on $\mathcal{F}_t$ and then on the auxiliary stage-I draw, and use $\|s_{\theta_t}\| \le B$:

$$\mathbb{E}[J(\theta_{t+1}) \mid \mathcal{F}_t] \ge J(\theta_t) + \alpha_t \|\nabla J(\theta_t)\|^2 + \alpha_t \mathbb{E}[\langle \nabla J(\theta_t), b_t \rangle \mid \mathcal{F}_t] - \tfrac{L_J B^2 \alpha_t^2}{2} \tag{87}$$

$$\ge J(\theta_t) + \tfrac{\alpha_t}{2} \|\nabla J(\theta_t)\|^2 - \tfrac{\alpha_t}{2} \mathbb{E}[\|b_t\|^2 \mid \mathcal{F}_t] - \tfrac{L_J B^2 \alpha_t^2}{2}, \tag{88}$$

where the second line uses the elementary Young inequality $\langle u, v \rangle \ge -\|u\|^2/2 - \|v\|^2/2$. Because $J \le 0$, the nonnegative process $-J(\theta_t)$ satisfies a Robbins–Siegmund inequality with summable error terms by (86) and (A4). Thus $\sum_t \alpha_t \|\nabla J(\theta_t)\|^2 < \infty$ almost surely, proving (81).

For completeness, this weighted-summability conclusion plus bounded increments implies actual norm convergence rather than merely a vanishing liminf. Set $u_t = \|\nabla J(\theta_t)\|$. The L_J-Lipschitz gradient and bounded actor score give $|u_{t+1} - u_t| \le L_J B \alpha_t$. If $u_t \ge \varepsilon > 0$ at arbitrarily late times, either $u_t \ge \varepsilon/2$ from some time onward (contradicting $\sum \alpha_t u_t^2 < \infty$ and $\sum \alpha_t = \infty$), or infinitely many disjoint excursions from below $\varepsilon/2$ to at least ε occur. Every such excursion has $\sum \alpha_t \ge \varepsilon/(2L_J B)$ and contributes at least $\varepsilon^3/(8L_J B)$ to $\sum \alpha_t u_t^2$, again a contradiction. Hence $u_t \to 0$ almost surely. Continuity of ∇J and boundedness of the iterates imply every accumulation point is stationary. □

F Additional audit of the gradient formulation

Exact conditional sampling is statistically well defined, but can be costly. Given an ideal feedback sample $f^\star$, repeatedly draw $X \sim p_\theta(\cdot \mid D)$ and $F \sim F_D(\cdot \mid X)$ until $F = f^\star$. The accepted X has law $r_\theta(\cdot \mid D, f^\star)$ because one-trial acceptance probability for $X = x$ equals $p_\theta(x \mid D) F_D(f^\star \mid x)$, whose sum is $m_\theta(f^\star \mid D)$. The conditional expected trials are $1/m_\theta(f^\star \mid D)$, so the average cost is

$$\mathbb{E}_{D \sim q^\star,\, f^\star \sim G_D^\star} \frac{1}{m_\theta(f^\star \mid D)}.$$

This gives an exact unbiased gradient implementation *using only the sampling resources postulated by GFT*, but offers no polynomial rollout guarantee and is especially problematic with raw, high-entropy feedback strings.

A finite-time approximate-gradient statement. For an actor-only implementation with fresh conditional samples from an auxiliary model of average TV error δ_t, assume J is L-smooth along actor steps, the trajectory score is bounded by B, and $\theta_{t+1} = \theta_t + \alpha s_{\theta_t}(X_t^+, D_t)$ for constant $\alpha > 0$. Repeating the conditional-bias bound and smooth-ascent argument above, without needing the auxiliary tracking theorem, yields

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}\|\nabla J(\theta_t)\|^2 \le \frac{2[-J(\theta_0)]}{\alpha T} + \frac{4B^2}{T} \sum_{t=0}^{T-1} \mathbb{E}\delta_t^2 + L\alpha B^2. \tag{89}$$

This is a *conditional finite-iteration stationarity bound*, not a bound on how many feedback examples are needed to achieve a given δ_t. Exact conditional sampling and $\alpha \asymp T^{-1/2}$ give an $O(T^{-1/2})$ average squared-gradient bound, not global optimality.

Why this does not yet solve the original sample-complexity question. Our new online theorem assumes $\ell(\cdot;\theta)$ is uniformly strongly convex and posterior-realizable in a parameter class with controlled score norms and bounded iterates. For tabular full-support conditional probability vectors these can hold when the parameter domain is appropriately bounded away from its boundary and contains the required posteriors; the constants and representation size may nevertheless depend exponentially on horizon. Natural-language partial auditors, unrestricted neural networks, off-policy replay, and jointly learned ideal-feedback generators are not covered. Without additional assumptions ruling out suboptimal stationary points, GFT's γ-separation translates convergence of *feedback distributions* to correctness, not convergence of neural gradients to correctness.